

# Chapter 1. Historical Perspective on the Development of Mechanical Ventilation

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## Historical Perspective on the Development of Mechanical Ventilation: Introduction

The history of mechanical ventilation is intimately intertwined with the history of anatomy, chemistry, and physiology; exploration under water and in the air; and of course, modern medicine. Anatomists described the structural connections of the lungs to the heart and vasculature and developed the earliest insights into the functional relationships of these organs. They emphasized the role of the lungs in bringing air into the body and probably expelling waste products, but showed little understanding of how air was used by the body. Chemists defined the constituents of air and explained the metabolic processes by which the cells used **oxygen** and produced carbon dioxide. Physiologists complemented these studies by exploring the relationships between levels of **oxygen** and carbon dioxide in the blood and ventilation. Explorers tested the true limits of physiology. Travel in the air and under water exposed humans to extremes in ventilatory demands and prompted the development of mechanical adjuncts to ventilation. Following the various historical threads provided by the anatomists, chemists, physiologists, and explorers provides a useful perspective on the tapestry of a technique modern physicians accept casually: mechanical ventilation.

## Anatomists of the Heart and Lungs

### Early Greeks

Early Greek physicians endorsed Empedocles' view that all matter was composed of four essential elements: earth, air, fire, and water. Each of these elements had primary qualities of heat, cold, moisture, and dryness.<sup>1</sup> Empedocles applied this global philosophic view to the human body by stating that "innate heat," or the soul, was distributed from the heart via the blood to various parts of the body.

The Hippocratic corpus stated that the purpose of respiration was to cool the heart. Air was thought to be pumped by the atria from the lungs to the right ventricle via the pulmonary artery and to the left ventricle through the pulmonary vein.<sup>2</sup> Aristotle believed that blood was an indispensable part of animals but that blood was found only in veins. Arteries, in contrast, contained only air. This conclusion probably was based on his methods of sacrificing animals. The animals were starved, to better define their vessels, and then strangled. During strangulation, blood pools in the right side of the heart and venous circulation, leaving the left side of the heart and arteries empty.<sup>2</sup> Aristotle described a three-chamber heart connected with passages leading in the direction of the lung, but these connections were minute and indiscernible.<sup>3</sup> Presumably, the lungs cooled the blood and somehow supplied it with air.<sup>4</sup>

Erasistratus (born around 300 BC) believed that air taken in by the lungs was transferred via the pulmonary artery to the left ventricle. Within the left ventricle, air was transformed into *pneuma zotikon*, or the "vital spirit," and was distributed through air-filled arteries to various parts of the body. The *pneuma zotikon* carried to the brain was secondarily changed to the *pneuma psychikon* ("animal spirit"). This animal spirit was transmitted to the muscles by the hollow nerves. Erasistratus understood that the right ventricle facilitated venous return by suction during diastole and that venous valves allowed only one-way flow of blood.<sup>1</sup>

The Greek physician Claudius Galen, practicing in Rome around AD 161, demonstrated that arteries contain blood by inserting a tube into the femoral artery of a dog.<sup>5,6</sup> Blood flow through the tube could be controlled by adjusting tension on a ligature placed around the proximal portion of the artery. He described a four-chamber heart with auricles distinct from the right and left ventricles. Galen also believed that the "power of pulsation has its origin in the heart itself" and that the "power [to contract and dilate] belongs by nature to the heart and is infused into the arteries from it."<sup>5,6</sup> He described valves in the heart and, as did Erasistratus, recognized their essential importance in preventing the backward discharge of blood from the heart. He alluded several times to blood flowing, for example, from the body through the vena cava into the right ventricle and even made the remarkable statement that "in the entire body the arteries come together with the veins and exchange air and blood through extremely fine invisible orifices."<sup>6</sup> Furthermore, Galen believed that "fuliginous wastes" were somehow discharged from the

blood through the lung.<sup>6</sup> Galen's appreciation that the lungs supplied some property of air to the body and discharged a waste product from the blood was the first true insight into the lung's role in ventilation. However, he failed in two critical ways to appreciate the true interaction of the heart and lungs. First, he believed, as did Aristotle and other earlier Greeks, that the left ventricle is the source of the innate heat that vitalizes the animal. Respiration in animals exists for the sake of the heart, which requires the substance of air to cool it. Expansion of the lung caused the lightest substance, that is, the outside air, to rush in and fill the bronchi. Galen provided no insight, though, into how air, or *pneuma*, might be drawn out from the bronchi and lungs into the heart. Second, he did not clearly describe the true circular nature of blood flow from the right ventricle through the lungs and into the left ventricle and then back to the right ventricle. His writings left the serious misconception that blood was somehow transported directly from the right to the left ventricle through the interventricular septum.<sup>1,5,6</sup>

## Renaissance Physicians

Byzantine and Arab scholars maintained Galen's legacy during the Dark Ages and provided a foundation for the rebirth of science during the Renaissance.<sup>1,6,7</sup> Around 1550, Vesalius corrected many inaccuracies in Galen's work and even questioned Galen's concept of blood flow from the right ventricle to the left ventricle. He was skeptical about the flow of blood through the interventricular pores Galen described.<sup>1,6,8</sup> Servetus, a fellow student of Vesalius in Paris, suggested that the vital spirit is elaborated both by the force of heat from the left ventricle and by a change in color of the blood to reddish yellow. This change in color "is generated in the lungs from a mixture of inspired air with elaborated subtle blood which the right ventricle of the heart communicates to the left. This communication, however, is made not through the middle wall of the heart, as is commonly believed, but by a very ingenious arrangement: the subtle blood courses through the lungs from the pulmonary artery to pulmonary vein, where it changes color. During this passage the blood is mixed with inspired air and through expiration it is cleansed of its sooty vapors. This mixture, suitably prepared for the production of the vital spirit, is drawn onward to the left ventricle of the heart by diastole."<sup>6,9</sup> Although Servetus' views proved ultimately to be correct, they were considered heretical at the time, and he was subsequently burned at the stake, along with most copies of his book, in 1553.

Columbus, a dissectionist to Vesalius at Padua, in 1559 suggested that blood travels to the lungs via the pulmonary artery and then, along with air, is taken to the left ventricle through the pulmonary vein. He further advanced the concept of circulation by noting that the left ventricle distributes blood to the body through the aorta, blood returns to the right ventricle in the vena cava, and venous valves in the heart allow only one-way flow.<sup>1,6,10</sup>

These views clearly influenced William Harvey, who studied anatomy with Fabricius in Padua from 1600 to 1602. Harvey set out to investigate the "true movement, pulse, action, use and usefulness of the heart and arteries." He questioned why the left ventricle and right ventricle traditionally were felt to play such fundamentally different roles. If the right ventricle existed simply to nourish the lungs, why was its structure so similar to that of the left ventricle? Furthermore, when one directly observed the beating heart in animals, it was clear that the function of both right and left ventricles also was similar. In both cases, when the ventricle contracted, it expelled blood, and when it relaxed, it received blood. Cardiac systole coincided with arterial pulsations. The motion of the auricles preceded that of the ventricles. Indeed, the motions are consecutive with a rhythm about them, the auricles contracting and forcing blood into the ventricles and the ventricles, in turn, contracting and forcing blood into the arteries. "Since blood is constantly sent from the right ventricle into the lungs through the pulmonary artery and likewise constantly is drawn the left ventricle from the lungs...it cannot do otherwise than flow through continuously. This flow must occur by way of tiny pores and vascular openings through the lungs. Thus, the right ventricle may be said to be made for the sake of transmitting blood through the lungs, not for nourishing them."<sup>6,11</sup>

Harvey described blood flow through the body as being circular. This was easily understood if one considered the quantity of blood pumped by the heart. If the heart pumped 1 to 2 drams of blood per beat and beat 1000 times per half-hour, it put out almost 2000 drams in this short time. This was more blood than was contained in the whole body. Clearly, the body could not produce amounts of blood fast enough to supply these needs. Where else could all the blood go but around and around "like a stage army in an opera." If this theory were correct, Harvey went on to say, then blood must be only a carrier of critical nutrients for the body. Presumably, the problem of the elimination of waste vapors from the lungs also was explained by the idea of blood as the carrier.<sup>1,6,11</sup>

With Harvey's remarkable insights, the relationship between the lungs and the heart and the role of blood were finally understood. Only two steps remained for the anatomists to resolve. First, the nature of the tiny pores and vascular openings through the lungs had to be explained. About 1650, Malpighi, working with early microscopes, found that air passes via the trachea and bronchi into and out of microscopic saccules with no clear connection to the bloodstream. He further described capillaries: "... and such is the wandering about of these vessels as they proceed on this side from the vein and on the other side from the artery, that the vessels no longer maintain a straight direction, but there appears a network made up of the articulations of the two vessels...blood flowed away along [these] tortuous vessels ... always contained within

tubules.”<sup>1,6</sup> Second, Borelli, a mathematician in Pisa and a friend of Malpighi, suggested the concept of diffusion. Air dissolved in liquids could pass through membranes without pores. Air and blood finally had been linked in a plausible manner.<sup>1</sup>

## Chemists and Physiologists of the Air and Blood

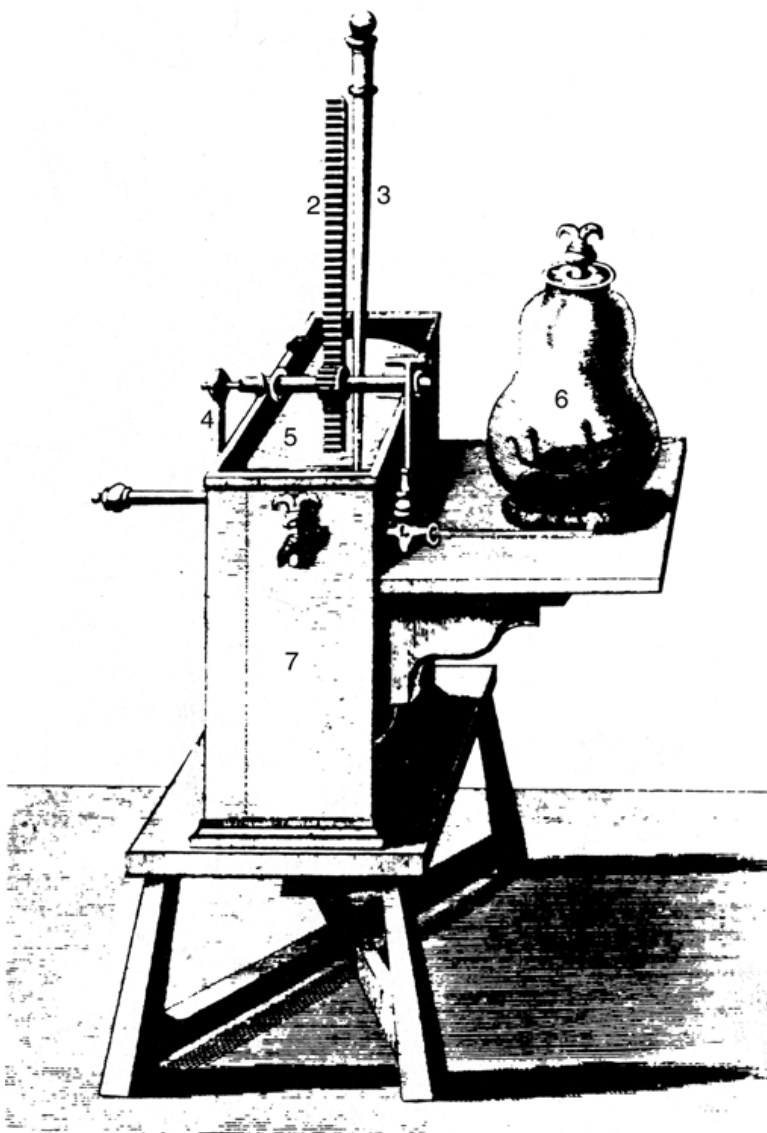
### Understanding Gases

The anatomists had identified an entirely new set of problems for chemists and physiologists to consider. The right ventricle pumped blood through the pulmonary artery to the lungs. In the lungs the blood took up some substance, evidenced by the change in color observed as blood passes through the pulmonary circulation. Presumably the blood released “fuliginous wastes” into the lung. The site of this exchange was thought to be at the alveolar–capillary interface, and it probably occurred by the process of diffusion. What were the substances exchanged between blood and air in the lung? What changed the color of blood and was essential for the production of the “innate heat”? What was the process by which “innate heat” was produced, and where did this combustion occur, in the left ventricle as supposed from the earliest Greek physician-philosophers or elsewhere? Where were the “fuliginous wastes” produced, and were they in any way related to the production of “innate heat”? If blood were a carrier, pumped by the left ventricle to the body, what was it carrying to the tissues and then again back to the heart?

Von Helmont, about 1620, added acid to limestone and potash and collected the “air” liberated by the chemical reaction. This “air” extinguished a flame and seemed to be similar to the gas produced by fermentation. This “air” also appeared to be the same gas as that found in the Grotto del Cane. This grotto was notorious for containing air that would kill dogs but spare their taller masters.<sup>1</sup> The gas, of course, was carbon dioxide. In the late seventeenth century, Boyle recognized that there is some substance in air that is necessary to keep a flame burning and an animal alive. Place a flame in a bell jar, and the flame eventually will go out. Place an animal in such a chamber, and the animal eventually will die. If another animal is placed in that same chamber soon thereafter, it will die suddenly. Mayow showed, around 1670, that enclosing a mouse in a bell jar resulted eventually in the mouse’s death. If the bell jar were covered by a moistened bladder, the bladder bulged inward when the mouse died. Obviously, the animals needed something in air for survival. Mayow called this the “nitro-aereal spirit,” and when it was depleted, the animals died.<sup>1,12</sup> This gas proved to be [oxygen](#). Boyle’s suspicions that air had other qualities primarily owing to its ingredients seemed well founded.<sup>13,14</sup>

In a remarkable and probably entirely intuitive insight, Mayow suggested that the ingredient essential for life, the “nitro-aereal spirit,” was taken up by the blood and formed the basis of muscular contraction. Evidence supporting this concept came indirectly. In the early 1600s, the concept of air pressure was first understood. von Guericke invented a pneumatic machine that reduced air pressure.<sup>1,15</sup> Robert Boyle later devised the pneumatic pump that could extract air from a closed vessel to produce something approaching a vacuum ([Fig. 1-1](#)). Boyle and Hooke used this pneumatic engine to study animals under low-pressure conditions. Apparently Hooke favored dramatic experiments, and he often demonstrated in front of crowds that small animals died after air was evacuated from the chamber. Hooke actually built a human-sized chamber in 1671 and volunteered to enter it. Fortunately, the pump effectively removed only about a quarter of the air, and Hooke survived.<sup>16</sup> Boyle believed that the difficulty encountered in breathing under these conditions was caused solely by the loss of elasticity in the air. He went on to observe, however, that animal blood bubbled when placed in a vacuum. This observation clearly showed that blood contained a gas of some type.<sup>13,14</sup> In 1727, Hales introduced the pneumatic trough ([Fig. 1-2](#)). With this device he was able to distinguish between free gas and gas no longer in its elastic state but combined with a liquid.<sup>1</sup> The basis for blood gas machines had been invented.

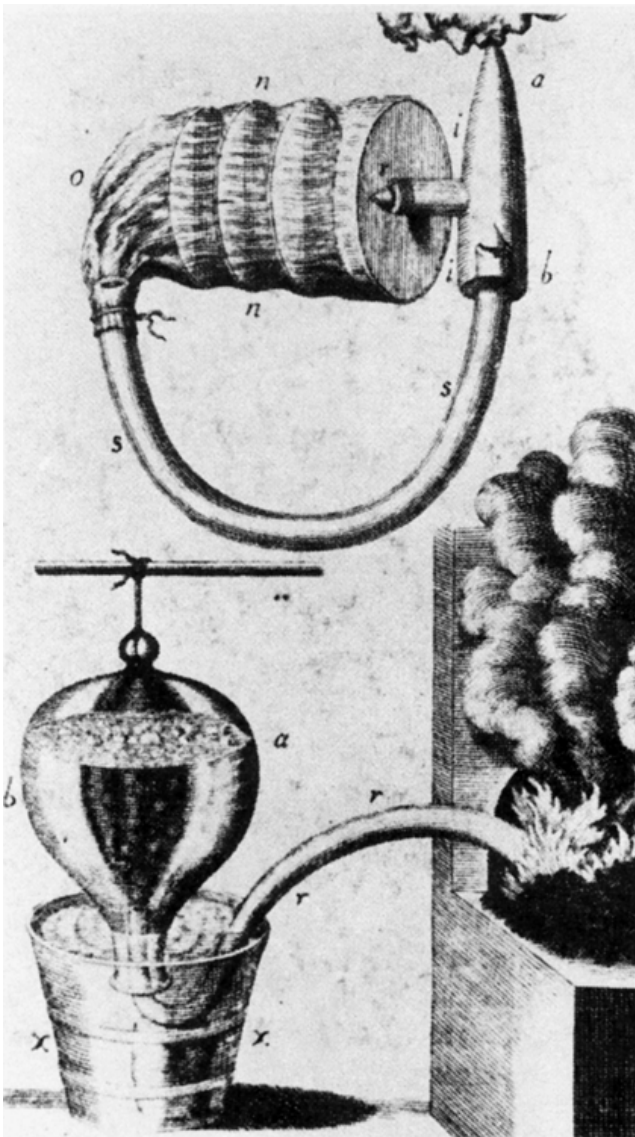
Figure 1-1



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A pneumatical engine, or vacuum pump, devised by Hooke in collaboration with Boyle around 1660. The jar (6) contains an animal in this illustration. Pressure is lowered in the jar by raising the tightly fitting slide (5) with the crank (4). (Used, with permission, from Graubard.<sup>6</sup>)

Figure 1-2



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In 1727, Hales developed the pneumatic trough, shown on the bottom of this illustration. This device enabled him to collect gases produced by heating. On the top is a closed-circuit respiratory apparatus for inhaling the collected gases. (Used, with permission, from Perkins.<sup>1</sup>)

The first constituent of air to be truly recognized was carbon dioxide. Joseph Black, around 1754, found that limestone was transformed into caustic lime and lost weight on being heated. The weight loss occurred because a gas was liberated during the heating process. The same results occurred when the carbonates of alkali metals were treated with an acid such as hydrochloric acid. He called the liberated gas “fixed air” and found that it would react with lime water to form a white insoluble precipitate of chalk. This reaction became an invaluable marker for the presence of “fixed air.” Black subsequently found that “fixed air” was produced by burning charcoal and fermenting beer. In a remarkable experiment he showed that “fixed air” was given off by respiration. In a Scottish church where a large congregation gathered for religious devotions, he allowed lime water to drip over rags in the air ducts. After the service, which lasted about 10 hours, he found a precipitate of crystalline lime ( $\text{CaCO}_3$ ) in the rags, proof that “fixed air” was produced during the services. Black recognized that “fixed air” was the same gas described by von Helmont that would extinguish flame and life.<sup>1,4,14</sup>

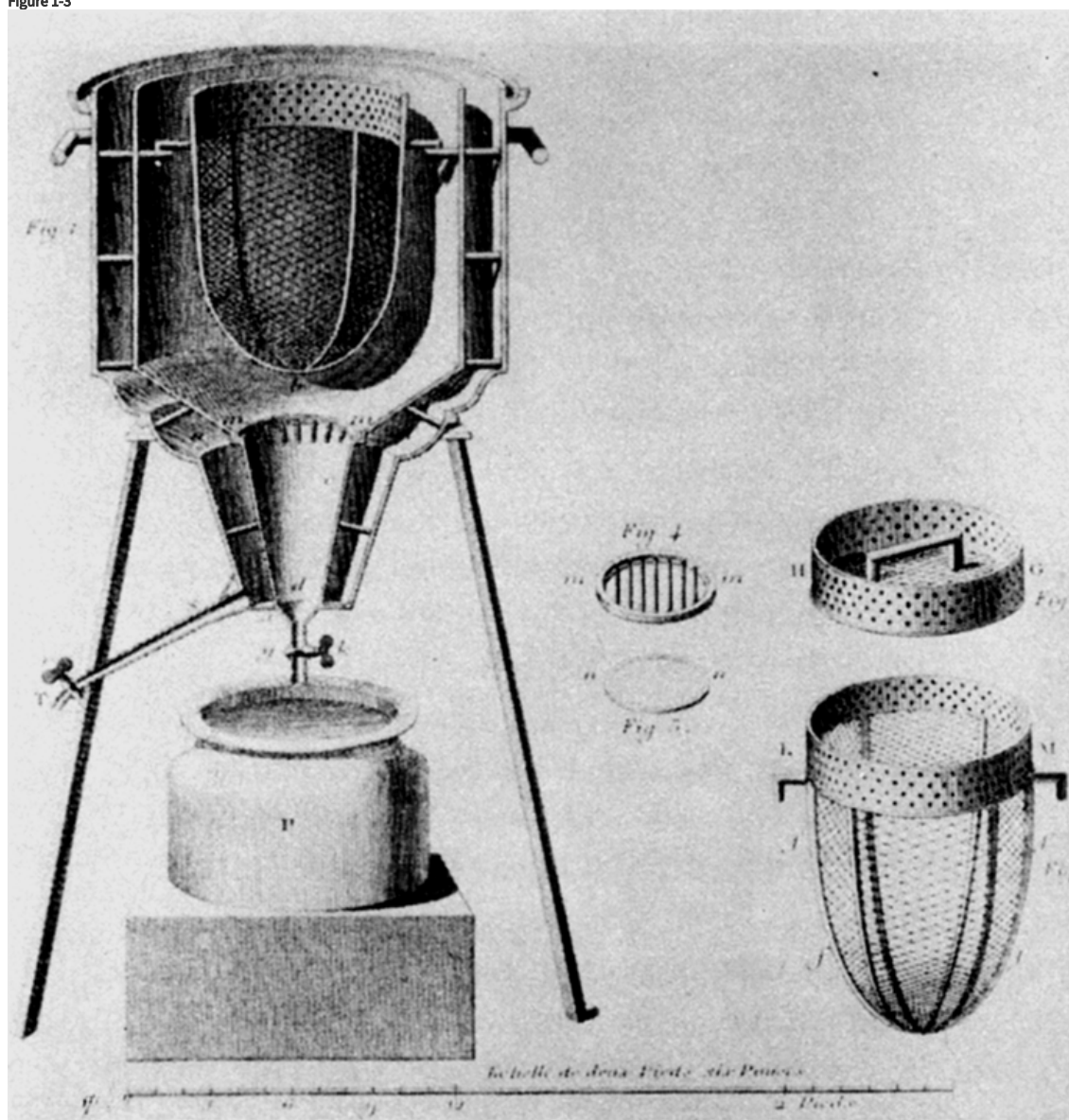
In the early 1770s, Priestley and Scheele, working independently of each other, both produced and isolated “pure air.” Priestley used a 12-inch lens to heat mercuric oxide. The gas released in this process passed through the long neck of a flask and was isolated over mercury. This gas allowed a flame to burn brighter and a mouse to live longer than in ordinary air.<sup>1,4,15</sup> Scheele also heated chemicals such as mercuric oxide and collected the gases in ox or hog bladders. Like Priestley, Scheele found that the gas isolated made a flame burn brighter. This gas was the “nitro-aereal spirit” described by Mayow. Priestley and Scheele described their observations to Antoine Lavoisier. He repeated Priestley’s experiments and found that if mercuric oxide was heated in the presence of charcoal, Black’s “fixed air” would be produced. Further work led Lavoisier to the conclusion that ordinary air must have at least two separate components. One part was respirable, combined with metals during heating, and

supported combustion. The other part was nonrespirable. In 1779, Lavoisier called the respirable component of air “**oxygen**.” He also concluded from his experiments that “fixed air” was a combination of coal and the respirable portion of air. Lavoisier realized that **oxygen** was the explanation for combustion.<sup>1,4,17</sup>

## Metabolism

In the 1780s, Lavoisier performed a brilliant series of studies with the French mathematician Laplace on the use of **oxygen** by animals. Lavoisier knew that **oxygen** was essential for combustion and necessary for life. Furthermore, he was well aware of the Greek concept of internal heat presumably produced by the left ventricle. The obvious question was whether animals used **oxygen** for some type of internal combustion. Would this internal combustion be similar to that readily perceived by the burning of coal? To answer this question, the two great scientists built an **ice calorimeter** (Fig. 1-3). This device could do two things. Because the melting **ice** consumed heat, the rate at which **ice** melted in the calorimeter could be used as a quantitative measure of heat production within the calorimeter. In addition, the consumption of **oxygen** could be measured. It then was a relatively simple task to put an animal inside the calorimeter and carefully measure heat production and **oxygen** consumption. As Lavoisier suspected, the amount of heat generated by the animal was similar to that produced by burning coal for the quantity of **oxygen** consumed.<sup>1,4</sup>

Figure 1-3



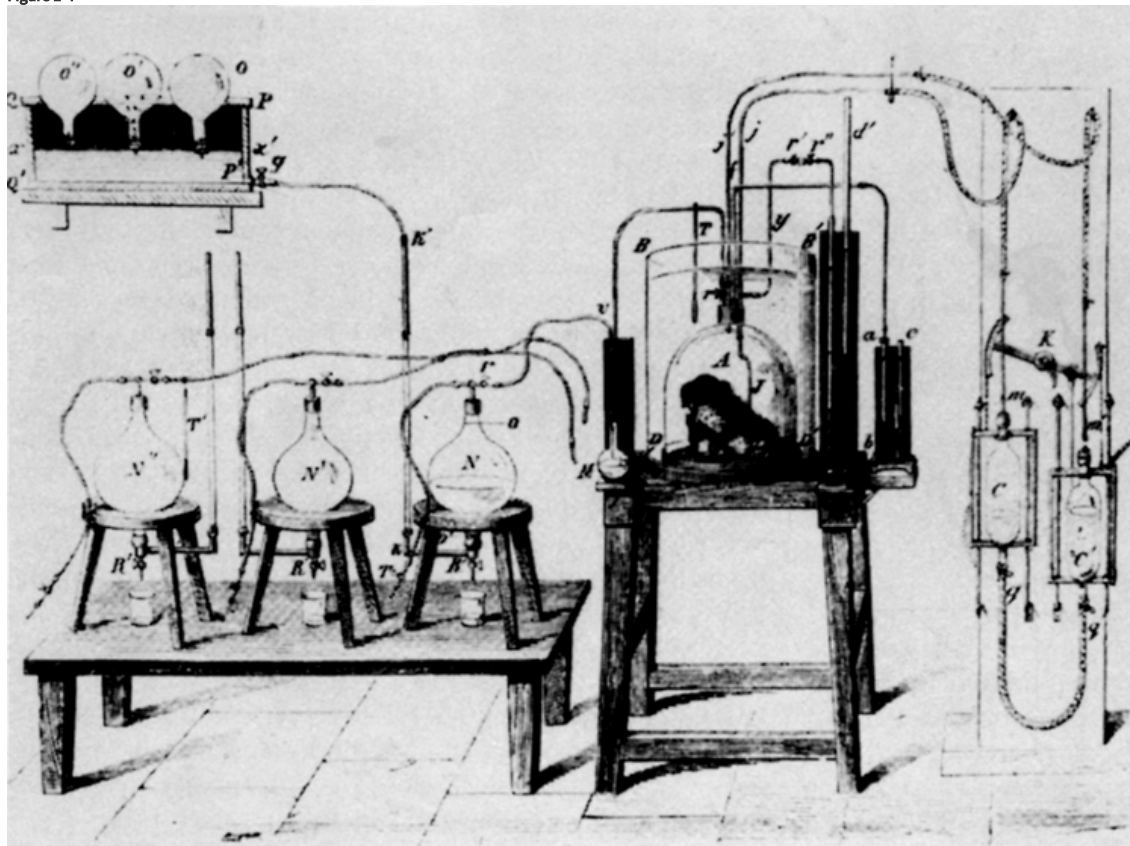
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The **ice** calorimeter, designed by Lavoisier and Laplace, allowed these French scientists to measure the **oxygen** consumed by an animal and the heat produced by that same animal. With careful measurements, the internal combustion of animals was found to be similar, in terms of **oxygen**

consumption and heat production, to open fires. (Used, with permission, from Perkins.<sup>1</sup>)

The Greeks suspected that the left ventricle produced innate heat, and Lavoisier himself may have thought that internal combustion occurred in the lungs.<sup>4</sup> Spallanzani, though, took a variety of tissues from freshly killed animals and found that they took up oxygen and released carbon dioxide.<sup>1</sup> Magnus, relying on improved methods of analyzing the gas content of blood, found higher oxygen levels in arterial blood than in venous blood but higher carbon dioxide levels in venous blood than in arterial blood. He believed that inhaled oxygen was absorbed into the blood, transported throughout the body, given off at the capillary level to the tissues, and there formed the basis for the formation of carbon dioxide.<sup>18</sup> In 1849, Regnault and Reiset perfected a closed-circuit metabolic chamber with devices for circulating air, absorbing carbon dioxide, and periodically adding oxygen (Fig. 1-4). Pettenkofer built a closed-circuit metabolic chamber large enough for a man and a bicycle ergometer (Fig. 1-5).<sup>19</sup> This device had a steam engine to pump air, gas meters to measure air volumes, and barium hydroxide to collect carbon dioxide. Although these devices were intended to examine the relationship between inhaled oxygen and exhaled carbon dioxide, they also could be viewed as among some of the earliest methods of controlled ventilation.

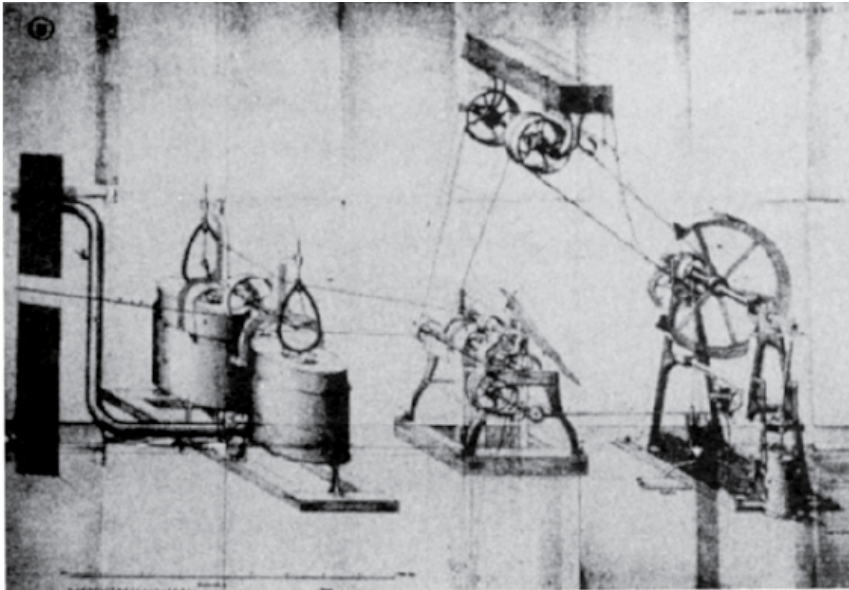
Figure 1-4



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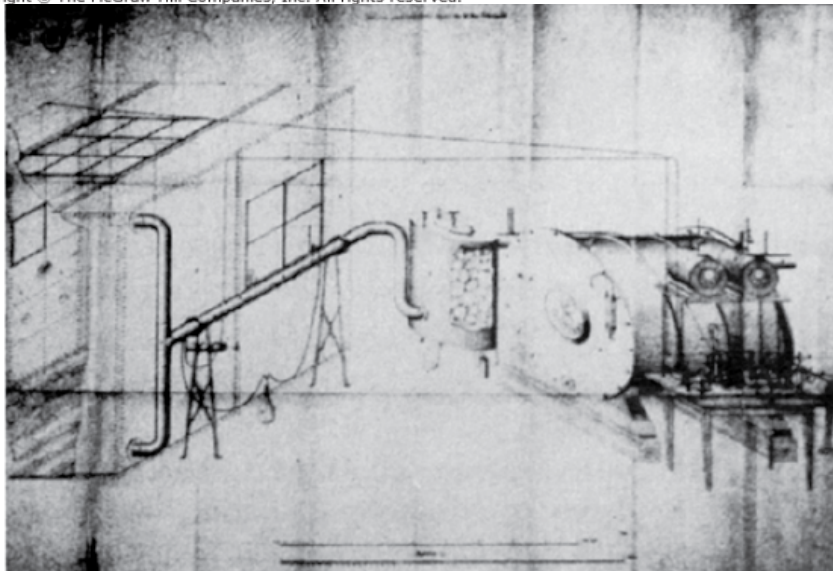
Regnault and Reiset developed a closed-circuit metabolic chamber in 1849 for studying oxygen consumption and carbon dioxide production in animals. (Used, with permission, from Perkins.<sup>1</sup>)

Figure 1-5



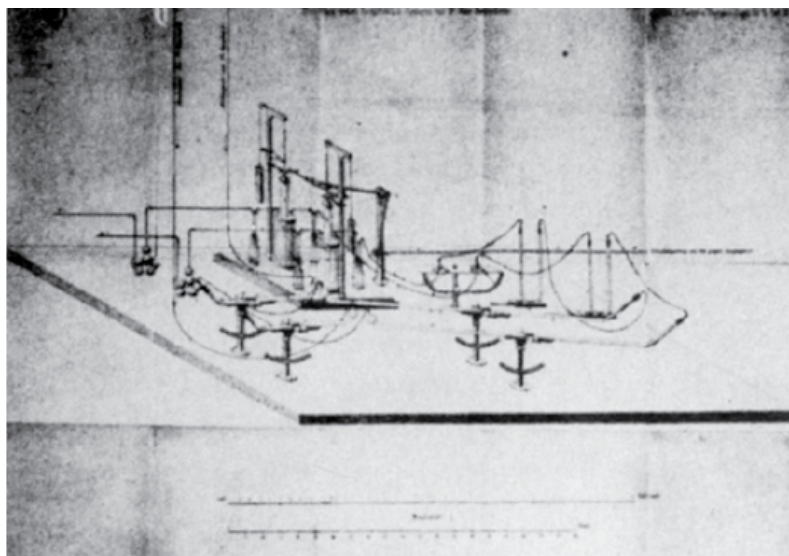
**A**

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**B**

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C

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**A.** This huge device, constructed by Pettenkoffer, was large enough for a person. **B.** The actual chamber. The gas meters used to measure gas volumes are shown next to the chamber. The steam engine and gasometers for circulating air are labeled A. **C.** A close-up view of the gas-absorbing device adjacent to the gas meter in **B.** With this device Pettenkoffer and Voit studied the effect of diet on the respiratory quotient. (Used, with permission, from Perkins.<sup>1</sup>)

## Blood Gases and Ventilation

In separate experiments, the British scientist Lower and the Irish scientist Boyle provided the first evidence that uptake of gases in the lungs was related to gas content in the blood. In 1669, Lower placed a cork in the trachea of an animal and found that arterial blood took on a venous appearance. Removing the cork and ventilating the lungs with a bellows made the arterial blood bright red again. Lower felt that the blood must take in air during its course through the lungs and therefore owed its bright color entirely to an admixture of air. Moreover, after the air had in large measure left the blood again in the viscera, the venous blood became dark red.<sup>20</sup> A year later Boyle showed with his vacuum pump that blood contained gas. Following Lavoisier's studies, scientists knew that **oxygen** was the component of air essential for life and that carbon dioxide was the "fuliginous waste."

About 1797, Davey measured the amount of **oxygen** and carbon dioxide extracted from blood by an air pump.<sup>4</sup> Magnus, in 1837, built a mercurial blood pump for quantitative analysis of blood **oxygen** and carbon dioxide content.<sup>18</sup> Blood was enclosed in a glass tube in continuity with a vacuum pump. Carbon dioxide extracted by means of the vacuum was quantified by the change in weight of carbon dioxide-absorbent caustic potash. **Oxygen** content was determined by detonating the gas in hydrogen.<sup>15</sup> A limiting factor in Magnus's work was the assumption that the quantity of **oxygen** and carbon dioxide in blood simply depended on absorption. Hence, the variables determining gas content in blood were presumed to be the absorption coefficients and partial pressures of the gases. In the 1860s, Meyer and Fernet showed that the gas content of blood was determined by more than just simple physical properties. Meyer found that the **oxygen** content of blood remained relatively stable despite large fluctuations in its partial pressure.<sup>21</sup> Fernet showed that blood absorbed more **oxygen** than did saline solution at a given partial pressure.<sup>15</sup>

Paul Bert proposed that **oxygen** consumption could not strictly depend on the physical properties of **oxygen** dissolving under pressure in the blood. As an example, he posed the problem of a bird in flight changing altitude abruptly. **Oxygen** consumption could be maintained with the sudden changes in pressure only if chemical reactions contributed to the oxygen-carrying capacity of blood.<sup>15</sup> In 1878, Bert described the curvilinear **oxygen** dissociation curves relating **oxygen** content of blood to its pressure. Hoppe-Seyler was instrumental in attributing the oxygen-carrying capacity of the blood to hemoglobin.<sup>22</sup> Besides his extensive experiments with animals in either high- or low-pressure chambers, Bert also examined the effect of ventilation on blood gas levels. Using a bellows to artificially ventilate animals through a tracheostomy, he found that increasing ventilation would increase **oxygen** content in blood and decrease the carbon dioxide content. Decreasing ventilation had the opposite effect.<sup>15</sup> Dohman, in Pflüger's laboratory, showed that both carbon dioxide excess and lack of **oxygen** would stimulate ventilation.<sup>23</sup> In 1885, Miescher-Rusch demonstrated that carbon dioxide excess was the more potent stimulus for ventilation.<sup>1</sup> Haldane and Priestley, building on this

work, made great strides in analyzing the chemical control of ventilation. They developed a device for sampling end-tidal, or alveolar exhaled, gas (Fig. 1-6). Even small changes in alveolar carbon dioxide fraction greatly increased minute ventilation, but hypoxia did not increase minute ventilation until the alveolar oxygen fraction fell to 12% to 13%.<sup>24</sup>

Figure 1-6



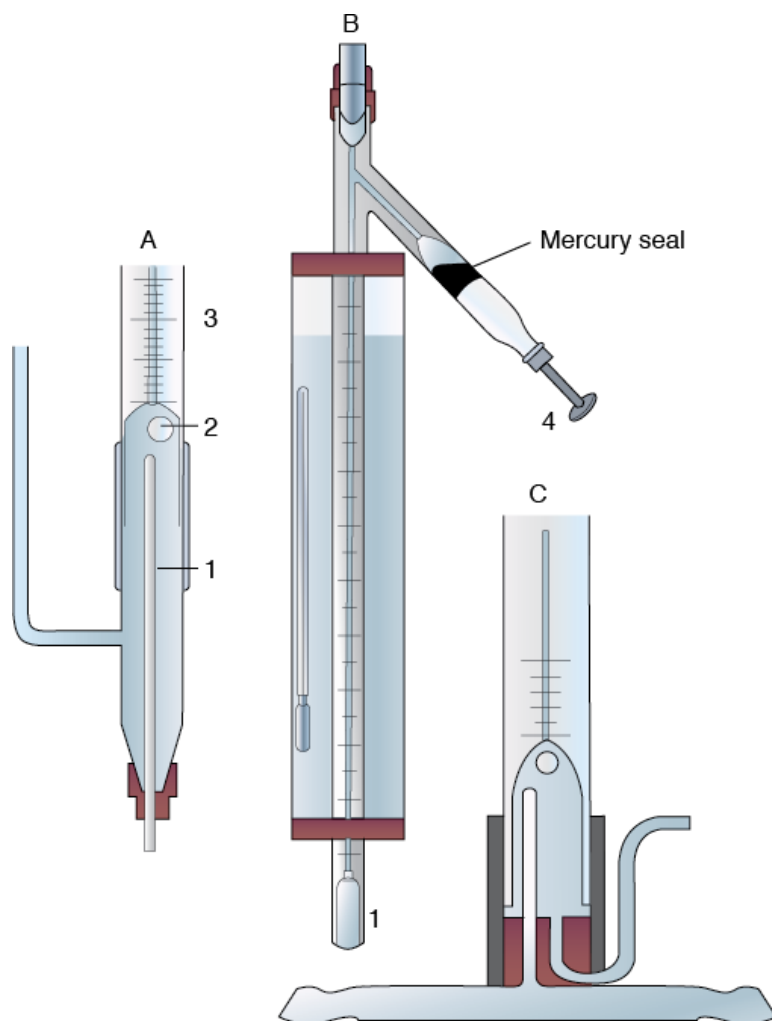
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This relatively simple device enabled Haldane and Priestley to collect end-tidal expired air, which they felt approximated alveolar air. The subject exhaled through the mouthpiece at the right. At the end of expiration, the stopcock on the accessory collecting bag was opened, and a small aliquot of air was trapped in this device. (Used, with permission, from Best CH, Taylor NB, *Physiological Basis of Medical Practice*. Baltimore, MD: Williams & Wilkins; 1939:509.)

Early measurements of arterial oxygen and carbon dioxide tensions led to widely divergent results. In Ludwig's laboratory the arterial partial pressure of oxygen was thought to be approximately 20 mm Hg. The partial pressure of carbon dioxide reportedly was much higher. These results could not entirely support the concept of passive gas movement between lung blood and tissues based on pressure gradients. Ludwig and others suspected that an active secretory process was involved in gas transport.<sup>4</sup> Coincidentally, the French biologist Biot observed that some deep-water fish had extremely large swim bladders. The gas composition in those swim bladders seemed to be different than that of atmospheric air. Biot concluded that gas was actively secreted into these bladders.<sup>4,15,24,25</sup> Pflüger and his coworkers developed the aerotonometer, a far more accurate device for measuring gas tensions than that used by Ludwig. When they obstructed a bronchus, they found no difference in the gas composition of air distal to the bronchial obstruction and that of pulmonary venous blood draining the area. They concluded that the lung did not rely on active processes for transporting oxygen and carbon dioxide; passive diffusion was a sufficient explanation.<sup>26</sup>

Although Pflüger's findings were fairly convincing at the time, Bohr resurrected this controversy.<sup>27</sup> He found greater variability in blood and air carbon dioxide and oxygen tensions than previously reported by Pflüger and suspected that under some circumstances secretion of gases might occur. In response, Krogh, a student of Bohr's, developed an improved blood gas-measuring technique relying on the microaeronometer (Fig. 1-7). With his wife, Krogh convincingly showed that alveolar air oxygen tension was higher than blood oxygen tension and vice versa for carbon dioxide tensions, even when the composition of inspired air was varied.<sup>28</sup> Douglas and Haldane confirmed Krogh's findings but wondered whether they were applicable only to people at rest. Perhaps during the stress of either exercise or high-altitude exposure, passive diffusion might not be sufficient. Indeed, the ability to secrete oxygen might explain the tolerance to high altitude developed by repeated or chronic exposures. Possibly carbon dioxide excretion might occur with increased carbon dioxide levels.<sup>26</sup> In a classic series of experiments, Marie Krogh showed that diffusion increased with exercise secondary to the concomitant increase in cardiac output.<sup>29</sup> Barcroft put to rest the diffusion-versus-secretion controversy with his "glass chamber" experiment. For 6 days he remained in a closed chamber subjected to hypoxia similar to that found on Pike's Peak. Oxygen saturation of radial artery blood was always less than that of blood exposed to simultaneously obtained alveolar gas, even during exercise. These were expected findings for gas transport based simply on passive diffusion.<sup>30</sup>

Figure 1-7



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Krogh's microaerometer. **A.** An enlarged view of the lower part of **B**. Through the bottom of the narrow tube (1) in **A**, blood is introduced. The blood leaves the upper end of the narrow tube (1) in a fine jet and plays on the air bubble (2). Once equilibrium is reached between the air bubble and blood, the air bubble is drawn by the screw plunger (4) into the graduated capillary tube shown in **B**. The volume of the air bubble is measured before and after treatment with KOH to absorb  $\text{CO}_2$  and potassium pyrogallate to absorb  $\text{O}_2$ . The changes in volume of the bubble reflect blood  $\text{CO}_2$  and  $\text{O}_2$  content. **C.** A model of **A** designed for direct connection to a blood vessel. (Used, with permission, from Best CH, Taylor NB, *Physiological Basis of Medical Practice*. Baltimore, MD: Williams & Wilkins; 1939:521.)

With this body of work, the chemists and physiologists had provided the fundamental knowledge necessary for the development of mechanical ventilation. **Oxygen** was the component of atmospheric gas understood to be essential for life. Carbon dioxide was the “fuliginous waste” gas released from the lungs. The exchange of **oxygen** and carbon dioxide between air and blood was determined by the tensions of these gases and simple passive diffusion. Blood was a carrier of these two gases, as Harvey first suggested. **Oxygen** was carried in two ways, both dissolved in plasma and chemically combined with hemoglobin. Blood carried **oxygen** to the tissues, where **oxygen** was used in cellular metabolism, that is, the production of the body’s “innate heat.” Carbon dioxide was the waste product of this reaction. **Oxygen** and carbon dioxide tensions in the blood were related to ventilation in two critical ways. Increasing ventilation would secondarily increase **oxygen** tensions and decrease carbon dioxide levels. Decreasing ventilation would have the opposite effect. Because blood levels of **oxygen** and carbon dioxide could be measured, physiologists now could assess the adequacy of ventilation. Decreased **oxygen** tensions and increased carbon dioxide tensions played a critical role in the chemical control of ventilation.

It was not understood, though, how carbon dioxide was carried by the blood until experiments performed by Bohr<sup>31</sup> and Haldane.<sup>32</sup> The concept of blood acid–base activity was just beginning to be examined in the early 1900s. By the 1930s, a practical electrode became available for determining anaerobic blood pH,<sup>33</sup> but pH was not thought to be useful clinically until the 1950s. In 1952, during the polio epidemic in Copenhagen, Ibsen suggested that hypoventilation, hypercapnia, and respiratory acidosis caused the high mortality rate in polio patients with respiratory paralysis. Clinicians disagreed because high blood levels of “bicarbonate” indicated an alkalosis. By measuring pH, Ibsen was proved

correct, and clinicians became acutely aware of the importance of determining both carbon dioxide levels and pH.<sup>4</sup> Numerous workers looked carefully at such factors as base excess, duration of hypercapnia, and renal buffering activity before Siggaard-Anderson published a pH/log  $P_{CO_2}$  acid-base chart in 1971.<sup>34</sup> This chart proved to be an invaluable basis for evaluating acute and chronic respiratory and metabolic acid-base disturbances. The development of practical blood gas machines suitable for use in clinical medicine did not occur until electrodes became available for measuring oxygen and carbon dioxide tensions in liquid solutions. Stow built the first electrode capable of measuring blood  $P_{CO_2}$ . As the basis for this device, he used a glass pH electrode with a coaxial central calomel electrode opening at its tip. A unique adaptation, however, was the use of a rubber finger cot to wrap the electrode. This wrap trapped a film of distilled water over the electrode. The finger cot then acted as a semipermeable membrane to separate the measuring electrode from the sample.<sup>35</sup> Clark used a similar idea in the development of an oxygen measuring device. Platinum electrodes were used as the measuring device, and polyethylene served as the semipermeable membrane.<sup>36</sup> By 1973, Radiometer was able to commercially produce the first automated blood gas analyzer, the ABL, capable of measuring  $P_{O_2}$ ,  $P_{CO_2}$ , and pH in blood.<sup>4</sup>

## Explorers and Working Men of Submarines and Balloons

Travel in the deep sea and flight have intrigued humankind for centuries. Achieving these goals has followed a typical pattern. First, individual explorers tested the limits of human endurance. As mechanical devices were developed to extend those limits, the deep sea and the air became accessible to commercial and military exploration. These forces further intensified the need for safe and efficient underwater and high-altitude travel. Unfortunately, the development of vehicles to carry humans aloft and under water proceeded faster than the appreciation of the physiologic risks. Calamitous events ensued, with serious injury and death often a consequence. Only a clear understanding of the ventilatory problems associated with flight and deep-sea travel has enabled human beings to reach outer space and the depths of the ocean floor.

### Exploration under Water

Diving bells undoubtedly were derived from ancient humans' inverting a clay pot over their heads and breathing the trapped air while under water. These devices were used in various forms by Alexander the Great at the siege of Tyre in 332 BC, the Romans in numerous naval battles, and pirates in the Black Sea.<sup>37,38</sup> In the 1500s, Sturmius constructed a heavy bell that, even though full of air, sank of its own weight. When the bell was positioned at the bottom of fairly shallow bodies of water, workers were able to enter and work within the protected area. Unfortunately, these bells had to be raised periodically to the surface to refresh the air. Although the nature of the foul air was not understood, an important principle of underwater work, the absolute need for adequate ventilation, was appreciated.<sup>15</sup>

Halley devised the first modern version of the diving bell in 1690 (Fig. 1-8). To drive out the air accumulated in the bell and "made foul" by the workers' respiration, small barrels of air were let down periodically from the surface and opened within the bell. Old air was released through the top of the bell by a valve. In 1691, Papin developed a technique for constantly injecting fresh air from the surface directly into the bell by means of a strong leather bellows. In 1788, Smeaton replaced the bellows with a pump for supplying fresh air to the submerged bell.<sup>15,37,38</sup>

Figure 1-8



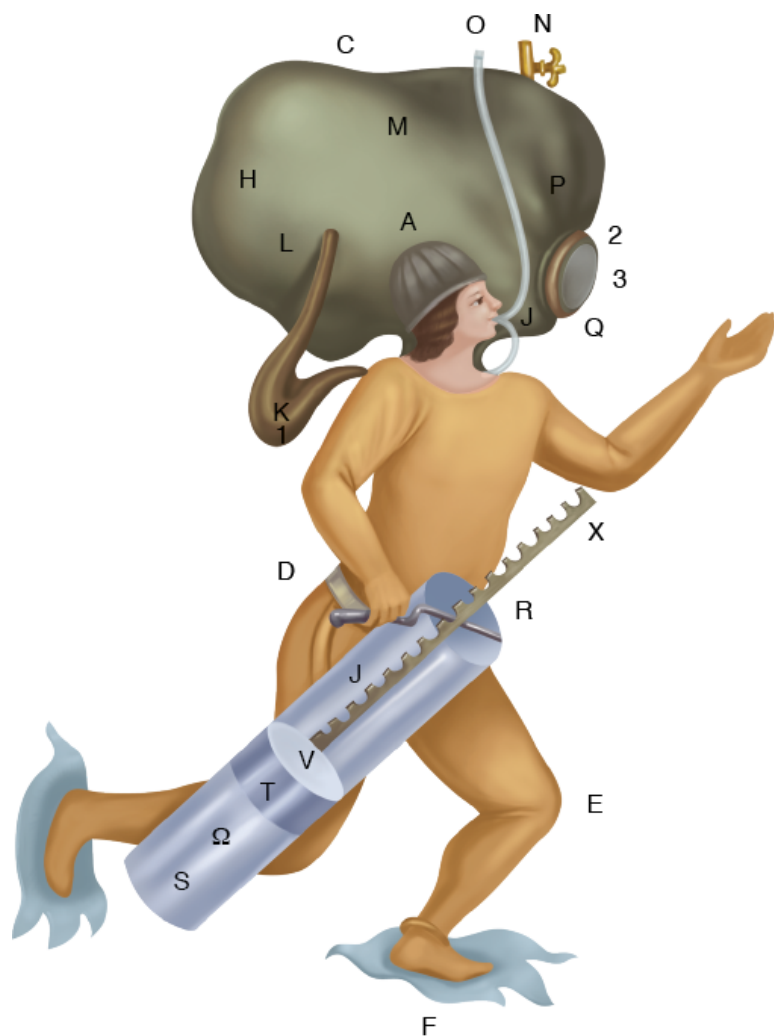
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Halley's version of the diving bell. Small barrels of fresh air were lowered periodically to the bell, and the worker inside the bell released the air. "Foul air" often was released by way of a valve at the top of the bell. Workers could exit the bell for short periods. (Used, with permission, from Hill.<sup>38</sup>)

Techniques used to make diving bells practical also were applied to divers. Xerxes used them to recover sunken treasure.<sup>39</sup> Sponge divers in the Mediterranean in the 1860s could stay submerged for 2 to 4 minutes and reach depths of 45 to 55 m.<sup>40</sup> Amas, female Japanese divers using only goggles and a weight to facilitate rapid descent, made dives to similar depths.<sup>41</sup> Despite the remarkable adaptations of breath-holding measures developed by these naked divers,<sup>42</sup> the commercial and military use of naked divers was limited. In 77 AD, Pliny described divers breathing through tubes while submerged and engaged in warfare. More sophisticated diving suits with breathing tubes were described by Leonardo da Vinci in 1500 and Renatus in 1511. Although these breathing tubes prolonged underwater activities, they did not enable divers to reach even moderate depths.<sup>15</sup> Borelli described a complete diving dress with tubes in the helmet for recirculating and purifying air in 1680 (Fig. 1-9).<sup>37</sup>

Figure 1-9

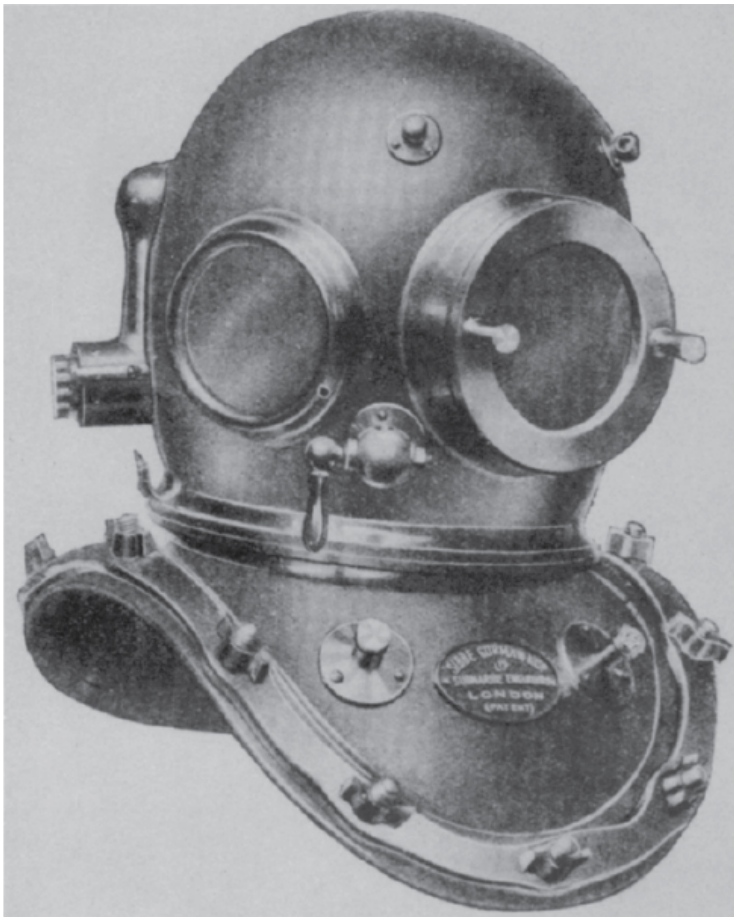


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A fanciful diving suit designed by Borelli in 1680. (Used, with permission, from Hill.<sup>38</sup>)

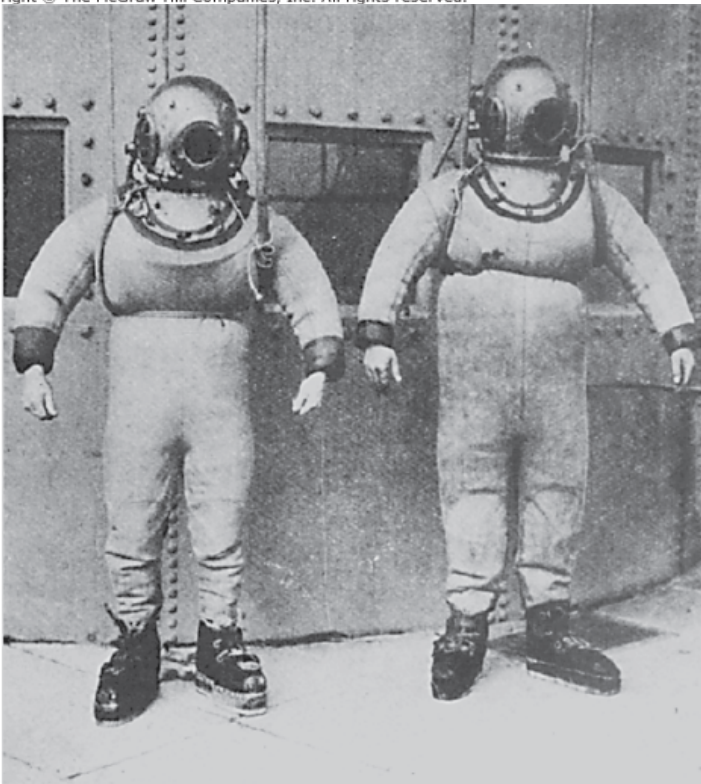
Klingert described the first modern diving suit in 1797.<sup>37</sup> It consisted of a large helmet connected by twin breathing pipes to an air reservoir that was large enough to have an associated platform. The diver stood on the platform and inhaled from the air reservoir through an intake pipe on the top of the reservoir and exhaled through a tube connected to the bottom of the reservoir. Siebe made the first commercially viable diving dress. The diver wore a metal helmet riveted to a flexible waterproof jacket. This jacket extended to the diver's waist but was not sealed. Air under pressure was pumped from the surface into the diver's helmet and escaped through the lower end of the jacket. In 1837, Siebe modified this diving dress by extending the jacket to cover the whole body. The suit was watertight at the wrists and ankles. Air under pressure entered the suit through a one-way valve at the back of the helmet and was released from the suit by an adjustable valve at the side of the helmet (Fig. 1-10).<sup>37</sup> In 1866, Denayouze incorporated a metal air reservoir on the back of the diver's suit. Air was pumped directly into the reservoir, and escape of air from the suit was adjusted by the diver.<sup>15</sup>

Figure 1-10



**A**

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**B**

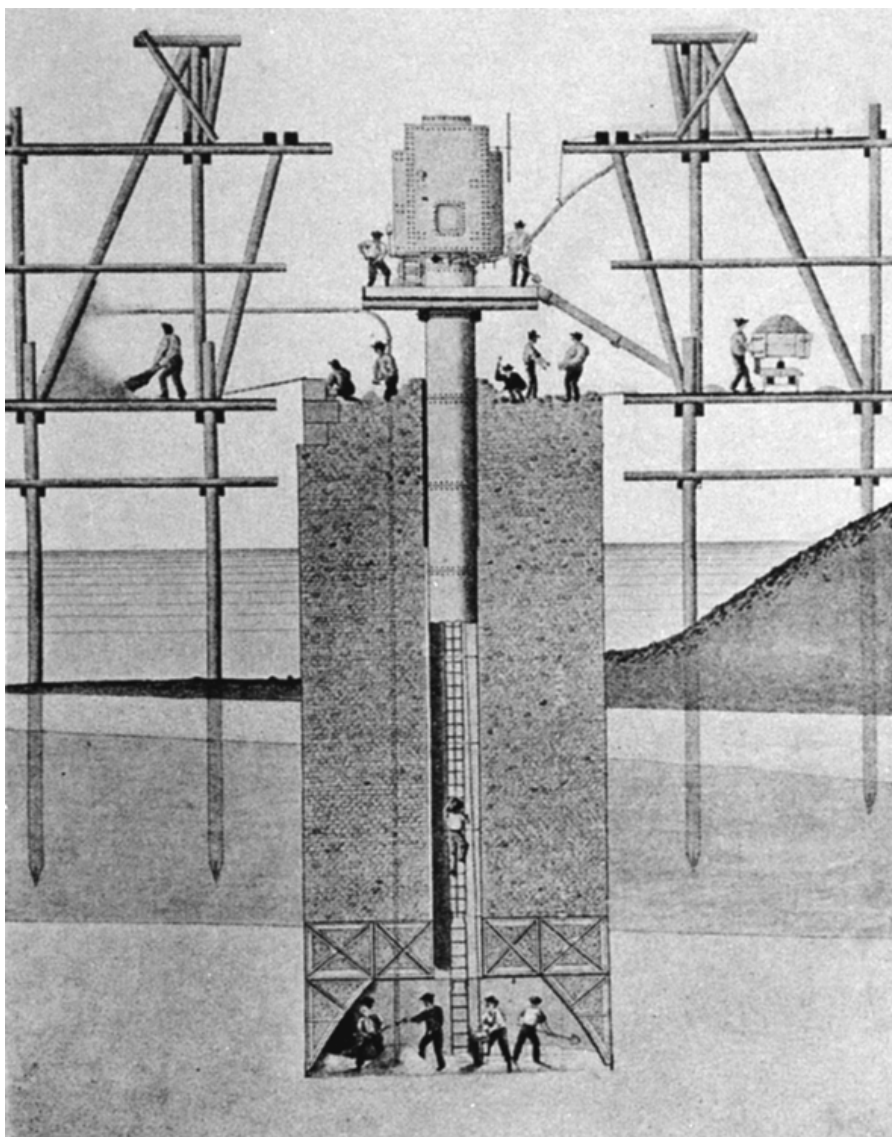
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**A.** The metal helmet devised by Siebe is still used today. **B.** The complete diving suit produced by Siebe, Gorman, and Company in the nineteenth century included the metal helmet, a diving dress sealed at the wrists and ankles, and weighted shoes. (Used, with permission, from Hill.<sup>38</sup>)

Siebe, Gorman, and Company produced the first practical self-contained diving dress in 1878. This suit had a copper chamber containing potash for absorbing carbon dioxide and a cylinder of oxygen under pressure.<sup>37</sup> Fleuss cleverly revised this diving suit in 1879 to include an oronasal mask with an inlet and an exhaust valve. The inlet valve allowed inspiration from a metal chamber containing oxygen under pressure. Expiration through the exhaust valve was directed into metal chambers under a breastplate that contained carbon dioxide absorbents. Construction of this appliance was so precise that Fleuss used it not only to stay under water for hours but also to enter chambers containing noxious gases. The Fleuss appliance was adapted rapidly and successfully to mine rescue work, where explosions and toxic gases previously had prevented such efforts.<sup>43</sup>

As Siebe, Gorman, and Company successfully marketed diving suits, commercial divers began to dive deeper and longer. Unfortunately, complications developed for two separate reasons. Decompression illness was recognized first. In 1830, Lord Cochrane took out a patent in England for “an apparatus for compressing atmospheric air within the interior capacity of subterraneous excavations [to]... counteract the tendency of superincumbent water to flow by gravitation into such excavations...and which apparatus at the same time is adapted to allowing workmen to carry out their ordinary operations of excavating, sinking, and mining.”<sup>38</sup> In 1841, Triger described the first practically applied caisson for penetrating the quicksands of the Loire River (Fig. 1-11).<sup>44</sup> This caisson, or hollow iron tube, was sunk to a depth of 20 m. The air within the caisson was compressed by a pump at the surface. The high air pressure within the caisson was sufficient to keep water out of the tube and allow workers to excavate the bottom. Once the excavation reached the prescribed depth, the caisson was filled with cement, providing a firm foundation. During the excavation process, workers entered and exited the caisson through an airlock. During this work, Triger described the first cases of “caisson disease,” or decompression illness, in workers after they had left the pressurized caisson. As this new technology was applied increasingly in shaft and tunnel work (e.g., the Douchy mines in France in 1846; bridges across the Midway and Tamar rivers in England in 1851 and 1855, respectively; and the Brooklyn Bridge, constructed between 1870 and 1873), caisson disease was recognized more frequently. Bert was especially instrumental in pointing out the dangers of high pressure.<sup>15</sup> Denayouze supervised many commercial divers and probably was among the first to recognize that decompression caused illness in these divers.<sup>15</sup> In the early 1900s, Haldane developed safe and acceptable techniques for staged decompression based on physiologic principles.<sup>38</sup>

Figure 1-11



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The caisson is a complex device enabling workers to function in dry conditions under shallow bodies of water or in other potentially flooded circumstances. A tube composed of concentric rings opens at the bottom to a widened chamber, where workers can be seen. At the top of the tube is a blowing chamber for maintaining air pressure and dry conditions within the tube. Workers enter at the top through an air lock and gain access to the working area via a ladder through the middle of the tube. (Used, with permission, from Hill.<sup>38</sup>)

Haldane also played a critical role in examining how well Siebe's closed diving suit supplied the ventilation needs of divers. This work may have been prompted by Bert's studies with animals placed in high-pressure chambers. Bert found that death invariably occurred when inspired carbon dioxide levels reached a certain threshold. Carbon dioxide absorbents placed in the high-pressure chamber prevented deaths.<sup>15</sup> Haldane's studies in this area were encouraged by a British Admiralty committee studying the risks of deep diving in 1906. Haldane understood that minute ventilation varied directly with alveolar carbon dioxide levels. It appeared reasonable that the same minute ventilation needed to maintain an appropriate  $PA_{CO_2}$  at sea level would be needed to maintain a similar  $PA_{CO_2}$  under water. What was not appreciated initially was that as the diver descended and pressure increased, pump ventilation at the surface necessarily also would have to increase to maintain minute ventilation. Haldane realized that at 2 atmospheres of pressure, or 33 ft under water, pump ventilation would have to double to ensure appropriate ventilation. This does not take into account muscular effort, which would further increase ventilatory demands. Unfortunately, early divers did not appreciate the need to adjust ventilation to the diving suit. Furthermore, air pumps often leaked or were maintained inadequately. Haldane demonstrated the relationship between divers' symptoms and hypercapnia by collecting exhaled gas from divers at various depths. The fraction of carbon dioxide in the divers' helmets ranged from 0.0018 to 0.10 atm.<sup>26,38</sup> The investigations of Bert and Haldane finally clarified the nature of "foul air" in diving bells and suits and the role of adequate ventilation in protecting underwater workers from hypercapnia. Besides his work with

divers, Haldane also demonstrated that the “black damp” found in mines actually was a dangerously toxic blend of 10% CO<sub>2</sub> and 1.45% O<sub>2</sub>.<sup>45</sup> He developed a self-contained rescue apparatus for use in mine accidents that apparently was more successful than the Fleuss appliance.<sup>46</sup>

Diving boats were fancifully described by Marsenius, in 1638, and others. Only the boat designed by Debrell in 1648 appeared plausible because “besides the mechanical contrivances of his boat, he had a chemical liquor, the fumes of which, when the vessel containing it was unstopped, would speedily restore to the air, fouled by the respiration, such a portion of vital spirits as would make it again fit for that office.” Although the liquor was never identified, it undoubtedly was an alkali for absorbing carbon dioxide.<sup>38</sup> Payerne built a submarine for underwater excavation in 1844. Since 1850, the modern submarine has been developed primarily for military actions at sea.

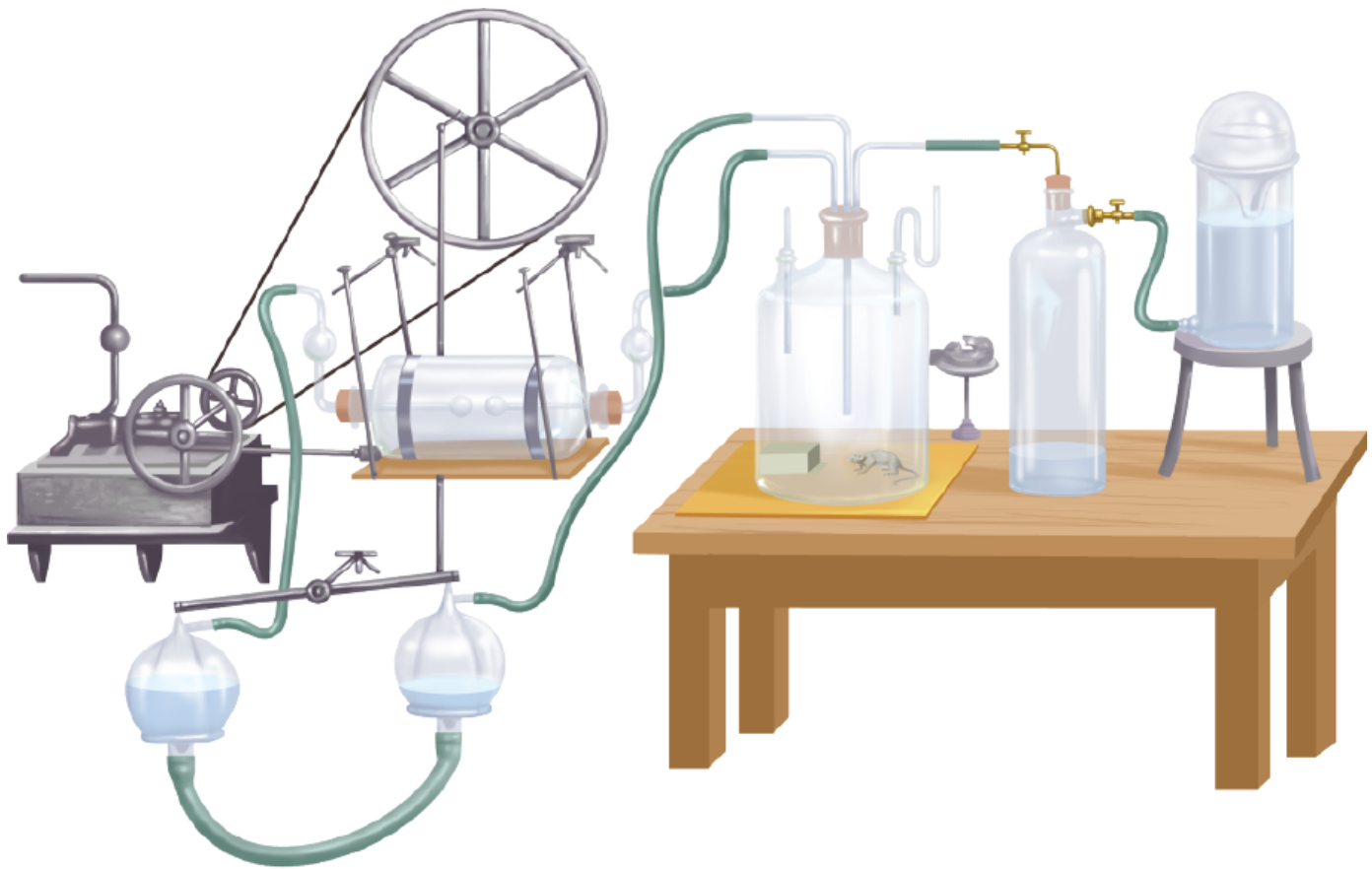
Submarines are an intriguing physiologic experiment in simultaneously ventilating many subjects. Ventilation in submarines is complex because it involves not only oxygen and carbon dioxide levels but also heat, humidity, and body odors. Early work in submarines documented substantial increases in temperature, humidity, and carbon dioxide levels.<sup>47</sup> Mechanical devices for absorption of carbon dioxide and air renewal were developed quickly,<sup>48</sup> and by 1928, Du Bois thought that submarines could remain submerged safely for up to 96 hours.<sup>49</sup> With the available carbon dioxide absorbents, such as caustic soda, caustic potash, and soda lime, carbon dioxide levels could be kept within relatively safe levels of less than 3%. Supplemental oxygen carried by the submarine could maintain a preferred fractional inspired oxygen concentration above 17%.<sup>39,50–52</sup>

## Exploration in the Air

In 1782, the Montgolfier brothers astounded the world by constructing a linen balloon about 18 m in diameter, filling it with hot air, and letting it rise about 2000 m into the air. On November 21, 1783, two Frenchmen, de Rozier and the Marquis d’Arlandes, were the first humans to fly in a Montgolfier balloon.<sup>53</sup> Within a few years, Jeffreys and Blanchard had crossed the English Channel in a balloon, and Charles had reached the astonishing height of 13,000 ft in a hydrogen-filled balloon. As with diving, however, the machines that carried them aloft brought human passengers past the limits of their physiologic endurance. Glaisher and Coxwell reached possibly 29,000 ft in 1862, but suffered temporary paralysis and loss of consciousness.<sup>4,26,54</sup>

Acoste’s description in 1573 of vomiting, disequilibrium, fatigue, and distressing grief as he traversed the Escaleras (Stairs) de Pariacaca, between Cuzco and Lima, Peru (“one of the highest places in the universe”), was widely known in Europe.<sup>55</sup> In 1804, von Humboldt attributed these high altitude symptoms to a lack of oxygen. Surprisingly, however, he found that the fraction of inspired oxygen in high-altitude air was similar to that found in sea-level air. He actually suggested that respiratory air might be used to prevent mountain sickness.<sup>56</sup> Longet expanded on this idea in 1857 by suggesting that the blood of high-altitude dwellers should have a lower oxygen content than that of sea-level natives. In a remarkable series of observations during the 1860s, Coindet described respiratory patterns of French people living at high altitude in Mexico City. Compared with sea-level values, respirations were deeper and more frequent, and the quantity of air expired in 1 minute was somewhat increased. He felt that “this is logical since the air of altitudes contains in a given volume less oxygen at a lower barometric pressure...[and therefore] a greater quantity of this air must be absorbed to compensate for the difference.”<sup>15</sup> Although these conclusions might seem reasonable now, physiologists of the time also considered decreased air elasticity, wind currents, exhalations from harmful plants, expansion of intestinal gas, and lack of support in blood vessels as other possible explanations for the breathing problems experienced at high altitude. Bert, the father of aviation medicine, was instrumental in clarifying the interrelationship among barometric pressure, oxygen tension, and symptoms. In experiments on animals exposed to low-pressure conditions in chambers (Fig. 1-12), carbonic acid levels increased within the chamber, but carbon dioxide absorbents did not prevent death. Supplemental oxygen, however, protected animals from dying under simulated high-altitude conditions (Fig. 1-13). More importantly, he recognized that death occurred as a result of the interaction of both the fraction of inspired oxygen and barometric pressure. When a multiple of these two variables—that is, the partial pressure of oxygen—reached a critical threshold, death ensued.<sup>15,39</sup>

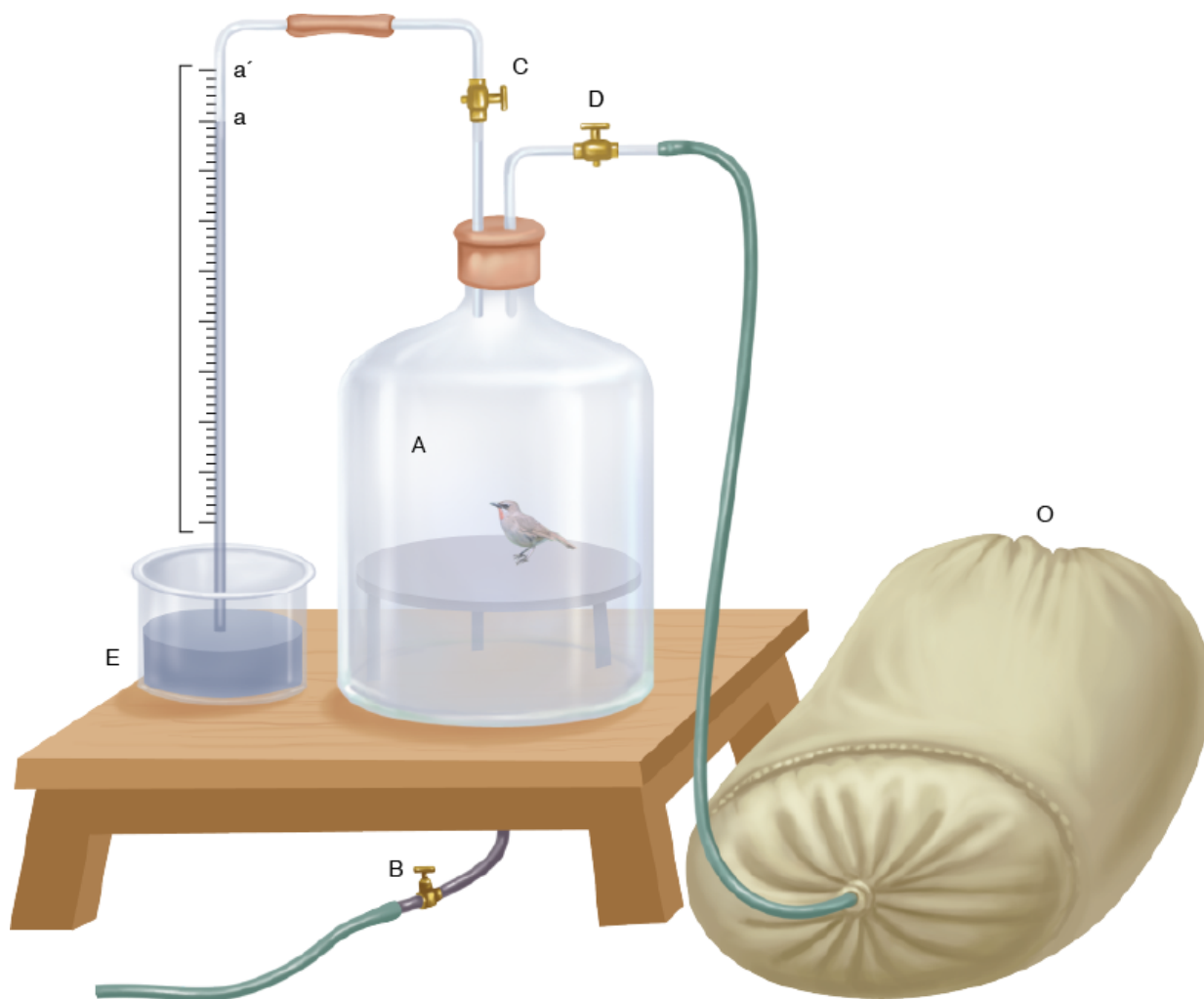
Figure 1-12



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A typical device used by Bert to study animals under low-pressure conditions. (Used, with permission, from Bert.<sup>15</sup>)

Figure 1-13

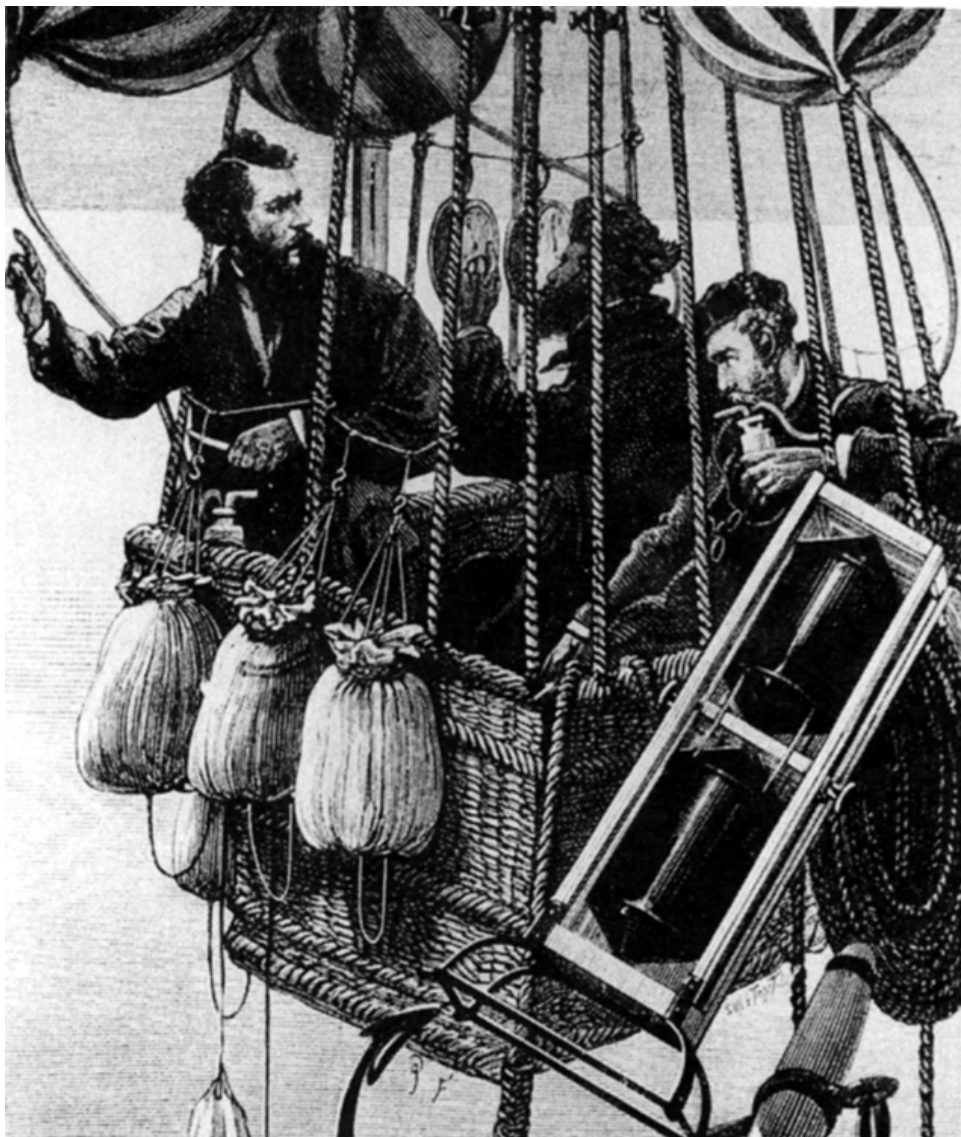


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A bird placed in a low-pressure bell jar can supplement the enclosed atmospheric air with **oxygen** inspired from the bag labeled O. Supplemental **oxygen** prolonged survival in these experiments. (Used, with permission, from Bert.<sup>15</sup>)

Croce-Spinelli, Sivel, and Tissandier were adventurous French balloonists eager to reach the record height of 8000 m. At Bert's urging, they experimented with the use of **oxygen** tanks in preliminary balloon flights and even in Bert's decompression chamber. In 1875, they began their historic attempt to set an altitude record supplied with **oxygen** cylinders (Fig. 1-14). Unfortunately, at 24,600 ft they released too much ballast, and their balloon ascended so rapidly that they were stricken unconscious before they could use the **oxygen**. When the balloon eventually returned to earth, only Tissandier remained alive.<sup>4,54</sup> This tragedy shook France. The idea that two men had died in the air was especially disquieting.<sup>53</sup> Unfortunately, the reasons for the deaths of Croce-Spinelli and Sivel were not clearly attributed to hypoxia. Von Schrotter, an Austrian physiologist, believed Bert's position regarding **oxygen** deficit as the lethal threat and encouraged Berson to attempt further high-altitude balloon flights. He originally devised a system for supplying **oxygen** from a steel cylinder with tubing leading to the balloonists. Later, von Schrotter conceived the idea of a face mask to supply **oxygen** more easily and also began to use liquid **oxygen**. With these devices, Berson reached 36,000 ft in 1901.<sup>4,54</sup>

Figure 1-14



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The adventurous French balloonists Croce-Spinelli, Sivel, and Tissandier begin their attempt at a record ascent. The balloonist at the right can be seen inhaling from an [oxygen](#) tank. Unfortunately, the supplemental [oxygen](#) did not prevent tragic results from a too rapid ascent. (Used, with permission, from Armstrong HG. *Principles and Practice of Aviation Medicine*. Baltimore, MD: Williams & Wilkins; 1939:4.)

The Wright brothers' historic flight at Kitty Hawk in 1903 substantially changed the nature of flight. The military value of airplanes soon was appreciated and applied during World War I. The Germans were especially interested in increasing the altitude limits for their pilots. They applied the concepts advocated by von Schrotter and provided liquid [oxygen](#) supplies for high-altitude bombing flights. Interest in airplane flights for commercial and military uses was especially high following Lindbergh's solo flight across the Atlantic in 1927. Much work was done on valves and [oxygen](#) gas regulators in the hope of further improving altitude tolerance. A series of high-altitude airplane flights using simple face masks and supplemental [oxygen](#) culminated in Donati's reaching an altitude of 47,358 ft in 1934. This was clearly the limit for human endurance using this technology.<sup>4,54</sup>

Somewhat before Donati's record, a breakthrough in flight was achieved by Piccard, who enclosed an aeronaut in a spherical metal chamber sealed with an ambient barometric pressure equivalent to that of sea level. The aeronaut easily exceeded Donati's record and reached 55,000 ft. This work recapitulated the important physiologic concept, gained from Bert's earlier experimental work in high-altitude chambers, that [oxygen](#) availability is a function of both fractional inspired [oxygen](#) and barometric pressure. Piccard's work stimulated two separate investigators to adapt pressurized diving suits for high-altitude flying. In 1933, Post devised a rubberized, hermetically sealed silk suit. In the same year, Ridge worked with Siebe, Gorman, and Company to modify a self-contained diving dress for flight. This suit provided [oxygen](#) under pressure and an air circulator with a soda lime canister for carbon dioxide removal. These suits proved quite successful, and soon pilots were exceeding heights of 50,000 ft. Parallel work with sealed gondolas attached to huge balloons led to ascents higher than 70,000 ft. In 1938, Lockheed produced the XC-

35, which was the first successful airplane with a pressurized cabin (Fig. 1-15).<sup>4,54</sup> These advances were applied quickly to military aviation in World War II. The German Air Ministry was particularly interested in developing oxygen regulators and valves and positive-pressure face masks for facilitating high-altitude flying.<sup>57</sup>

Figure 1-15



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Lockheed produced the XC-35 in 1938. This was the first plane to have a pressurized cabin. (Used, with permission, Armstrong HG. *Principles and Practice of Aviation Medicine*. Baltimore, MD: Williams & Wilkins; 1939:337.)

Work throughout World War II defined limits for technological support of high-altitude flight. Pilots could reach up to 12,000 ft safely without oxygen supplements. Above this limit, oxygen-enriched air was essential. With flights going above 25,000 ft, oxygen supplementation alone usually was insufficient, and some type of pressurized system—cabin, suit, or mask—was needed. Pressurization as an adjunct, however, reached its limit of usefulness at approximately 80,000 ft. At this altitude, air compressors became too leaky and inefficient to maintain adequate pressurization. A completely sealed cabin was essential to protect passengers adequately from the rarefied atmosphere outside. An altitude of 80,000 ft thus became a functional definition of space because at this height complete control of the atmosphere in the plane (i.e., the supply of oxygen, a means of removing carbon dioxide, and adequate control of temperature and humidity) was required.<sup>58,59</sup> Advances in submarine ventilatory physiology were adapted to the space program. In 1947, the American Air Force began the XI program, which culminated in the production in 1952 of the X15 aircraft. This plane reached a top speed of 4159 miles per hour at an altitude of 314,750 ft. More importantly, the technology developed for this plane was a prelude to manned satellite programs. The United States Mercury and the Russian Vostok programs both relied on rockets to boost small, one-person capsules into space orbit. The Mercury capsule had a pure oxygen atmosphere at a reduced cabin pressure. In addition, the pilot wore a pressurized suit with an independent, closed oxygen supply. In April 1961, Gagarin was the first person to be launched into space. Shepard followed soon after, in May 1961, and reached an altitude of 116 miles. More sophisticated space flight—in the Gemini, Apollo, and space station programs—was based on similar ventilation systems and principles.<sup>60</sup>

## Mechanical Ventilation of Resuscitation and Anesthesia

### Vivisection

Galen described ventilating an animal as follows: “If you take a dead animal and blow air through its larynx [through a reed], you will fill its bronchi and watch its lungs attain the greatest distention.”<sup>61</sup> Unfortunately, Galen failed to appreciate how ventilating the lungs could help him in his vivisection work. Galen operated on many living animals, but his studies on the function of the heart were limited by the risk of pneumothorax. Opening the thoracic cavity almost certainly resulted in death of the animal.<sup>1,6</sup> More than a thousand years later, Vesalius

realized that ventilation could protect animals from pneumothorax.<sup>62,63</sup> The lungs would collapse and the beating heart would almost stop when Vesalius opened the chest cavity, but the heart could be restarted by inflating the lungs through a reed tied into the trachea. Paracelsus, a contemporary of Vesalius, is reported to have used a similar technique around 1530 in attempting to resuscitate a human. Did Paracelsus adapt Vesalius's research efforts, or vice versa?<sup>63</sup> It is also unclear whether Vesalius himself tried artificial ventilation during the dissection of a Spanish nobleman. Legend has it that when the nobleman's heart began to beat once more, Vesalius's medical associates were so outraged that they reported him to the religious authorities. Vesalius only avoided being burned at the stake by embarking on a pilgrimage to the Holy Land, but he died during the voyage.<sup>64</sup>

Presumably, Harvey became familiar with Vesalius's use of ventilation during vivisection because he mentioned artificial ventilation in his work later in England.<sup>63</sup> Other English scientists soon after began to mention artificial ventilation in their own studies.<sup>65</sup> In 1664, Hooke dramatically described dissecting a dog, placing a pipe into the windpipe of the animal, and using a pair of bellows to ventilate the dog (and keep the heart beating) for longer than an hour.<sup>63</sup> Lower, an associate of Hooke, showed that artificial respiration kept the color of blood red during dissection.<sup>63</sup>

## Resuscitating the Apparently Drowned

Artificial respiration with a bellows and tracheal tube remained popular for vivisection work but was applied to humans only after a curious turn of events. Attempts to resuscitate apparently dead people were first recorded in the mid-eighteenth century. The origins of this movement are not entirely clear. Indeed, there were strong reasons for people to fear the dead. The risk of contagious disease was well known—memories of the plague were still fresh—and religious beliefs dissuaded many from believing in the wisdom of resuscitation. Despite these disincentives, sporadic attempts were made at organized resuscitation. In 1740, the Académie des Sciences in Paris issued an *avis* strongly advising mouth-to-mouth respiration for resuscitating the apparently drowned.<sup>63</sup> In 1744, Tossach used this technique successfully in saving a life.<sup>66</sup> Fothergill soon after provided an excellent description of the mouth-to-mouth resuscitation technique, including the use of bellows if the “blast of a man's mouth” was not sufficient.<sup>63,64,67,68</sup> In 1760, Buchan went on to advise creating “an opening in the windpipe” when air cannot be forced into the chest through the mouth or nose.<sup>69</sup> Societal pressures led to widespread dissemination of knowledge about resuscitation techniques. In response to citizens' concerns about the large number of lives lost in canals, a group of influential laymen in Amsterdam formed the Society for the Rescue of Drowned Persons (*Maatschappij tot Redding von Dreykhingen*) in 1767.<sup>63,64,67</sup> The express purpose of this society was to publicize the need for and the techniques of resuscitation. Similar societies soon were formed in other maritime cities, such as Venice and Milan in 1768, Paris in 1771, London in 1774, and Philadelphia in 1780.

The Dutch method emphasized five steps: keeping the patient warm, artificial respiration through the mouth, fumigation with tobacco smoke through the rectum (Fig. 1-16), stimulants placed orally or rectally, and bleeding. Cogan, an English physician with a Dutch wife, translated a pamphlet describing the Dutch method into English. Hawes, an apothecary, read the pamphlet and led a concerted effort to introduce this technique into England. In encouraging this work, Hawes' activities led directly to the formation of the Royal Humane Society in 1774.<sup>67</sup> Through this society, many physicians were encouraged to develop techniques for resuscitating the apparently drowned. In 1776, Hunter advocated the use of a double bellows for artificial ventilation. The first stroke blew fresh air into the lung, and the second stroke sucked out stale air. He had perfected this technique during physiologic studies with dogs. Hunter advised the use of Priestley's pure air (oxygen) for resuscitation, but it is unclear whether this advice was ever followed.<sup>64,67,70</sup> In 1776, Cullen suggested relying on tracheal intubation and bellows ventilation for reviving the apparently dead.<sup>71</sup> In 1791, Curry developed an intralaryngeal cannula for this purpose, as did Fine in 1800. These cannulas could be placed through the nose, mouth, or trachea.

Figure 1-16

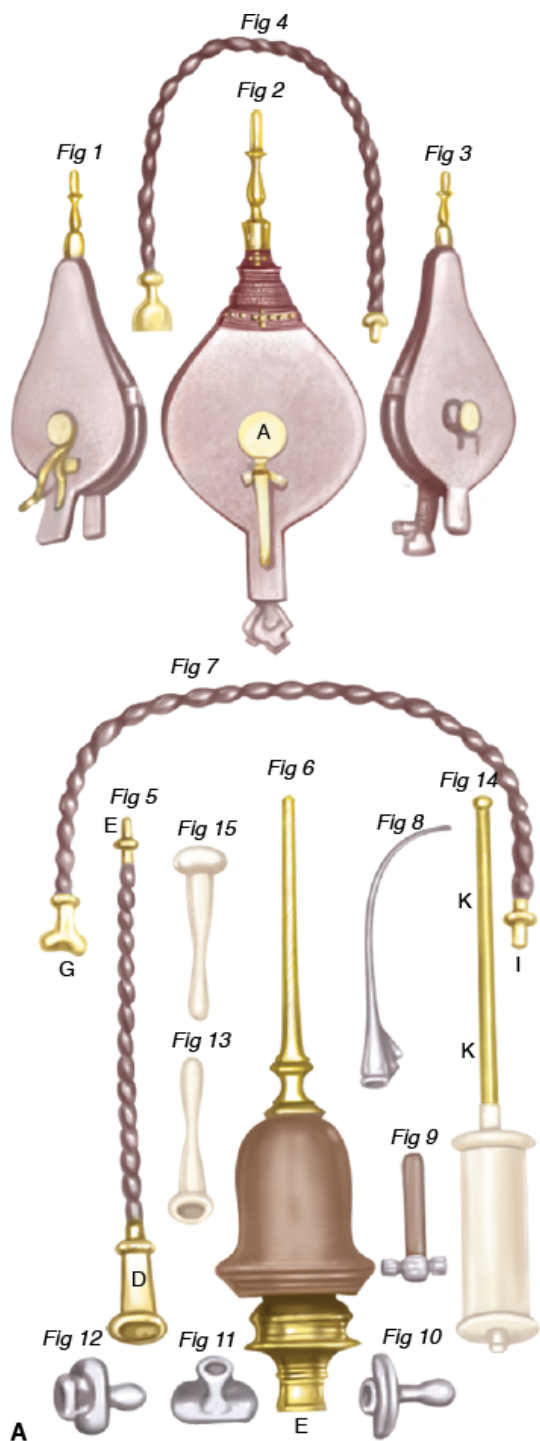


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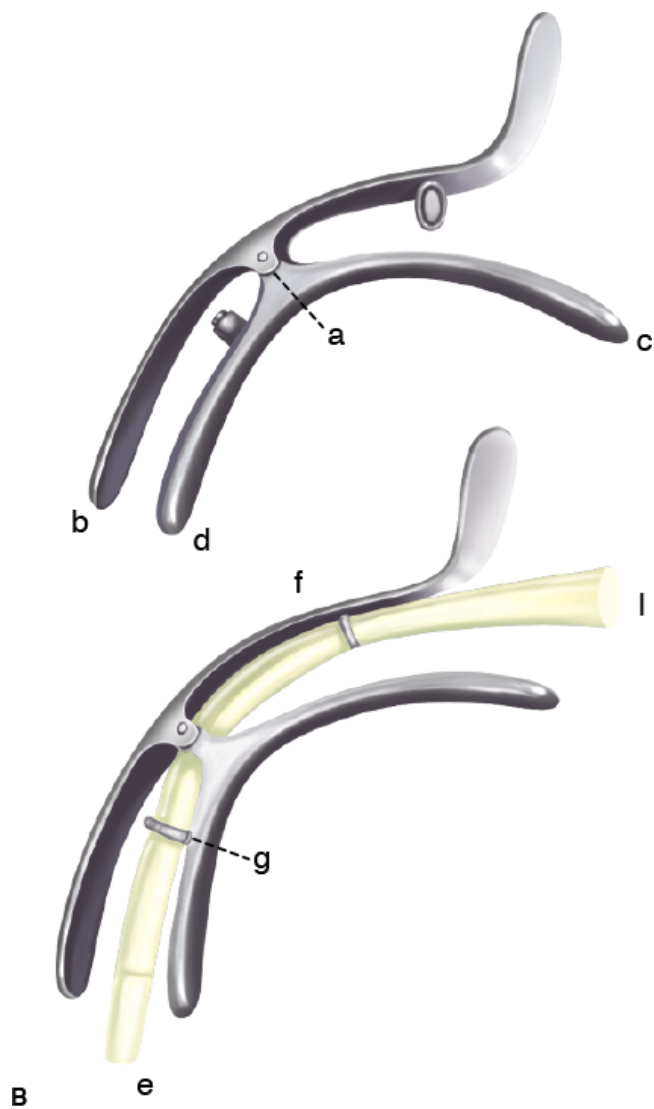
An attempt at resuscitating an apparently drowned person using the modified Dutch method. One resuscitator is assisting respiration by massaging the chest. The fumigator is instilling tobacco smoke through the rectum. (Used, with permission, from Morsch.<sup>64</sup>)

Many other physicians were encouraged to develop ingenious devices as resuscitation aids by the Royal Humane Society (Fig. 1-17). This society held competitions and offered prizes and medals for the best work in this area.<sup>63,64,67,72</sup> As an alternative to tracheal intubation, Chaussier constructed a simple bag and face mask for artificial ventilation in 1780 (Fig. 1-18). He thought that this device would protect the rescuer from the deleterious effects of exhaled air. Chaussier devised accessory tubing for the face mask to allow the use of supplemental oxygen.<sup>73</sup> Kite, Curry, and Chaussier also developed devices to assist the operator in cannulating the trachea through the mouth.<sup>73,74</sup>

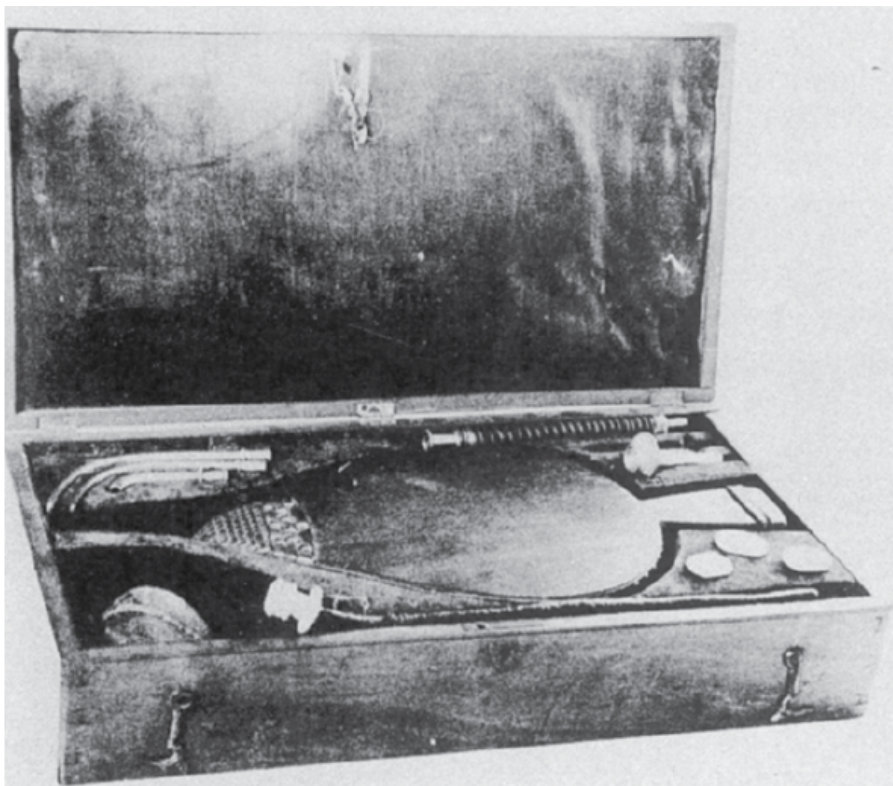
Figure 1-17



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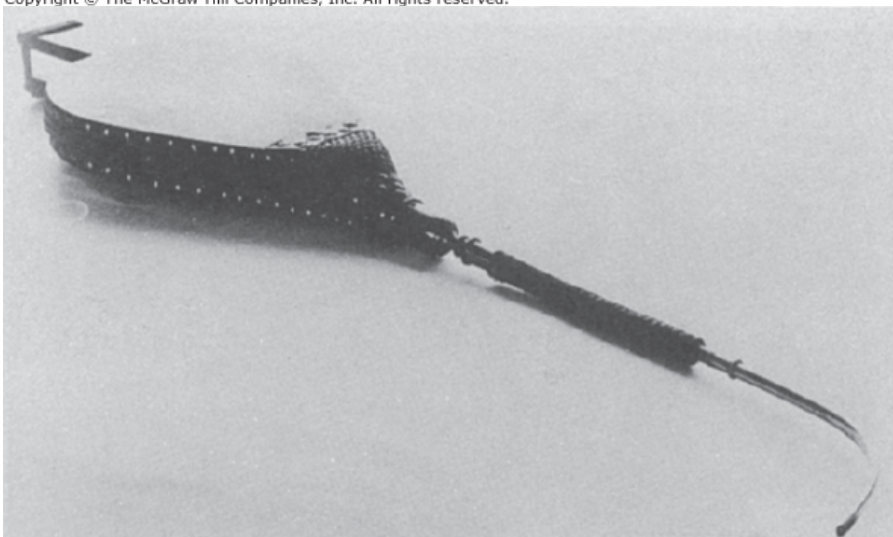


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**C**

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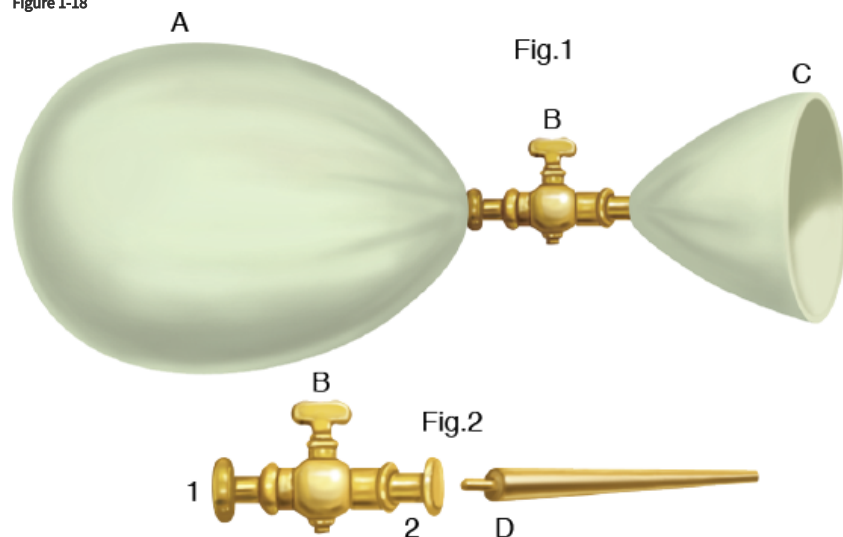


**D**

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**A.** Examples of some of the devices included in the Royal Humane Society's compendium of resuscitation techniques in 1806. Figures 1, 2, and 3 are bellows of different sizes. Figure 6 is a brass box for holding a stimulating substance. Various connecting tubes and nozzles are also enclosed. (Used, with permission, from Mushin WW, Rendell-Baker L, *The Principles of Thoracic Anesthesia*. Oxford, England: Blackwell Scientific Publications; 1953:32.) **B.** A two-bladed intubating spatula was developed to hold the mouth open and allow passage of the tracheal tube through the larynx. (Used, with permission, from Mushin WW, Rendell-Baker L, *The Principles of Thoracic Anesthesia*. Oxford, England: Blackwell Scientific Publications; 1953:36.) **C.** The Royal Humane Society approved this type of box of intubation equipment with a bellows ventilator for distribution in 1806. (Used, with permission, from McClellan I. Nineteenth century resuscitation apparatus. *Anaesthesia*. 1981;36(3):308.) **D.** The bellows is shown connected to the otolaryngeal cannula and ready for use. (Used, with permission, from McClellan I. Nineteenth century resuscitation apparatus. *Anaesthesia*. 1981;36(3):308.)

Figure 1-18



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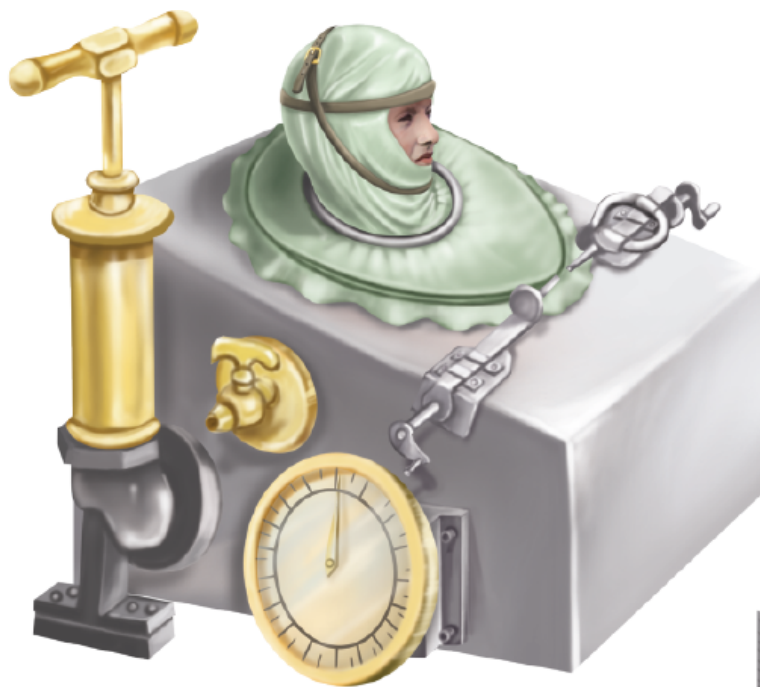
Chaussier developed this face mask and bag for artificial ventilation in 1780. (Used, with permission, from Mushin WW, Rendell-Baker L. *The Principles of Thoracic Anesthesia*. Oxford, England: Blackwell Scientific Publications; 1953:39.)

As these techniques for resuscitation were gaining widespread acceptance, concerns were being raised about the effectiveness of bellows ventilation. Leroy, in a dramatic series of studies in 1827 and 1828, subjected an animal to overzealous bellows inflation and caused fatal pneumothorax.<sup>75,76</sup> Although later it was realized that the pressures reached in this demonstration were unlikely to be achieved in clinical practice,<sup>63</sup> the French Academy quickly condemned the technique. Despite adaptations of bellows to limit ventilatory volumes,<sup>77</sup> the Royal Humane Society also abandoned the use of tracheal intubation and bellows ventilation for resuscitation.<sup>63</sup> Consequently, positive-pressure ventilation was banned from medical practice early in its infancy, not to be routinely relied on for patient care until well into the twentieth century.

## Negative-Pressure Ventilators

As an alternative to positive-pressure ventilation, physicians began to develop machines for negative-pressure ventilation. The first tank respirator was produced by Dalziel, of Scotland, in 1832. It was an airtight box in which the patient sat enclosed up to the neck. Negative pressure was created by bellows placed within the box but operated from the outside by a piston rod and one-way valve.<sup>78,79</sup> Jones, of Kentucky, patented the first tank respirator in America in 1864. The design appears similar to that of Dalziel's apparatus (Fig. 1-19).<sup>78,80</sup> Although Jones used this device to treat asthma and bronchitis, he also claimed cures for paralysis, neuralgia, rheumatism, seminal weakness, and dyspepsia.<sup>80</sup> Von Hauke designed a series of cuirass and tank respirators in the 1870s that were intended specifically to treat patients with respiratory diseases, but he showed little insight into the physiologic basis for how this type of respirator might be of benefit in lung disease. Woillez presented his version of a tank respirator to the French Academy of Medicine in 1876 (Fig. 1-20). It was basically a hollow cylinder of metal with a rigid lower end and an upper end enclosing a neck made of a rubber diaphragm seal. Air was evacuated from the cylinder by a bellows. Woillez understood the physiologic basis of ventilation and incorporated a bar placed on the patient's sternum to measure tidal excursions and adequacy of ventilation. Unfortunately, this device seemed to have been used only for resuscitating the apparently drowned and with little success.<sup>78,81</sup>

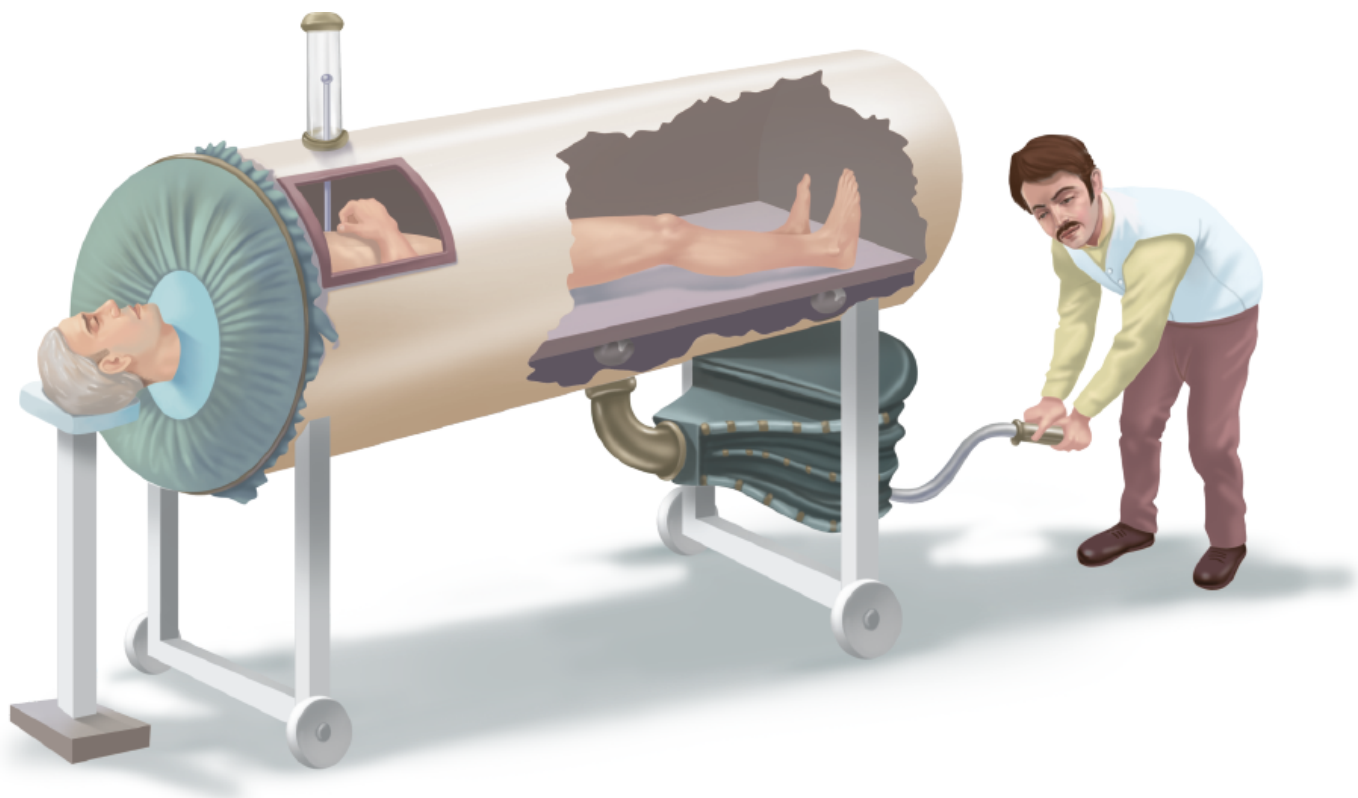
Figure 1-19



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The body-enclosing tank respirator constructed by Jones in 1864. The large syringe was used to create negative pressure. (Used, with permission, from Emerson JH. *Evolution of Iron Lungs*. Cambridge, MA: JH Emerson; 1978:fig. 1.)

Figure 1-20



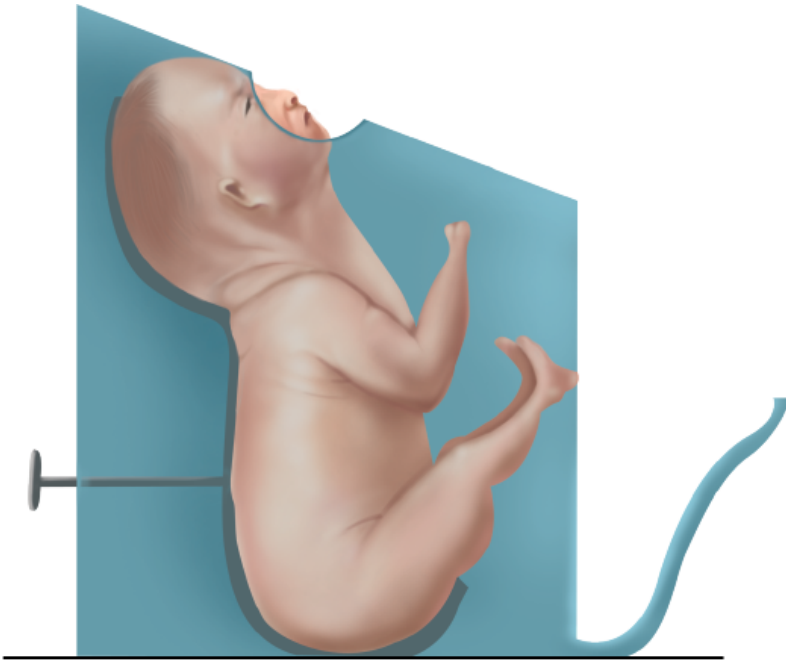
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The spirophore produced by Woillez in 1876 had a rod placed on the patient's sternum to indicate the adequacy of tidal excursions. (Used, with permission, from Emerson JH: *Evolution of Iron Lungs*. Cambridge, MA: JH Emerson; 1978:fig. 2.)

Many other ingenious devices were invented for negative-pressure ventilation over the next 50 years. Breuillard, of Paris, patented a tank respirator operated by a steam boiler.<sup>78,80</sup> Bell devised a vacuum jacket for newborns with neonatal respiratory distress. A resuscitation box developed by Braun reportedly was quite useful for children with respiratory distress (Fig. 1-21).<sup>78,80,82</sup> Eisenmenger designed a prototype respirator that extended from the upper part of the sternum to the pubis (Fig. 1-22A). A more sophisticated device covered the chest and only a portion of the abdomen (Fig. 1-22B). These devices allowed positive-pressure compression of the chest to assist exhalation and negative-pressure suction to facilitate inhalation.<sup>78,80,83</sup> A later device, called the *Eisenmenger biomotor*, was patented in 1927, and had only an abdominal cuirass shell.<sup>84</sup> An important advantage of these devices was the access they allowed to the patient for nursing care. This consideration prompted Lord, of Worcester, to build a respirator room (Fig. 1-23). Huge pistons in the ceiling created the pressure changes but required heavy equipment.<sup>78,80</sup> Severy, in 1916, and Schwake, in 1926, built negative-pressure ventilators that required the patient to stand (Fig. 1-24). Although they incorporated ingenious mechanical elements, their practicality for severely ill patients was limited.<sup>80</sup>

Figure 1-21

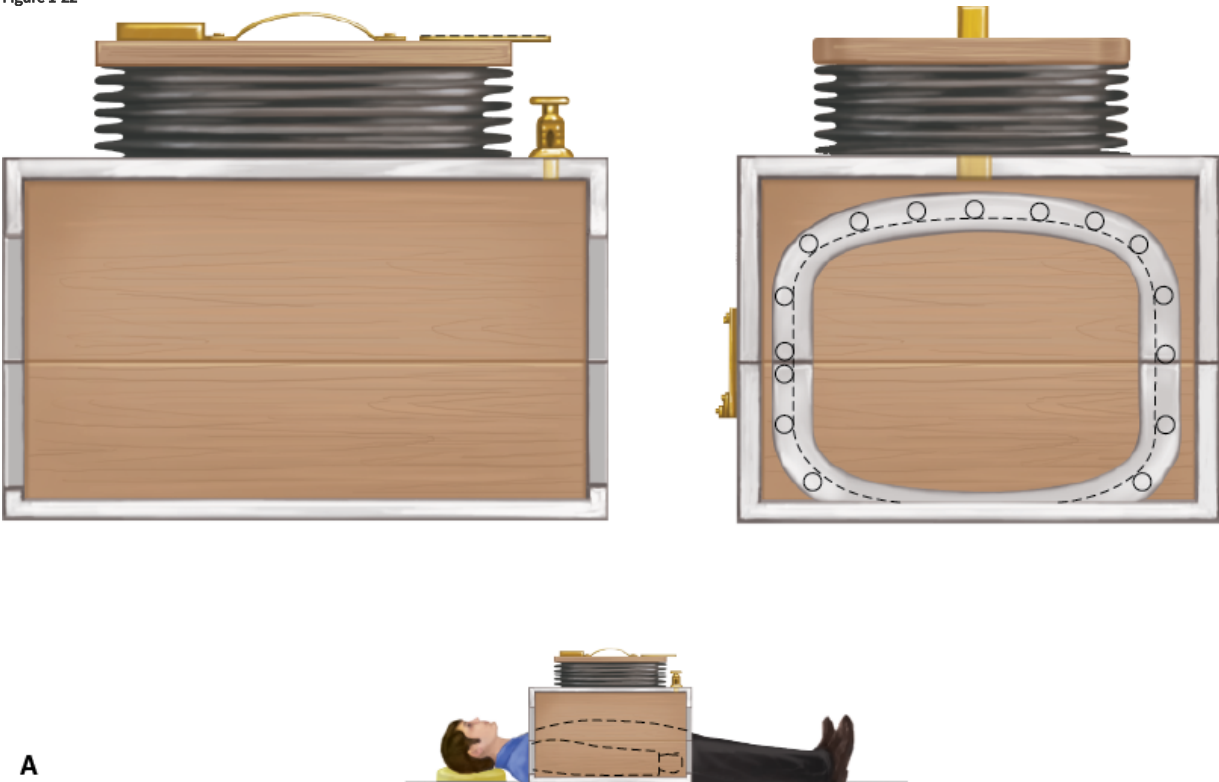


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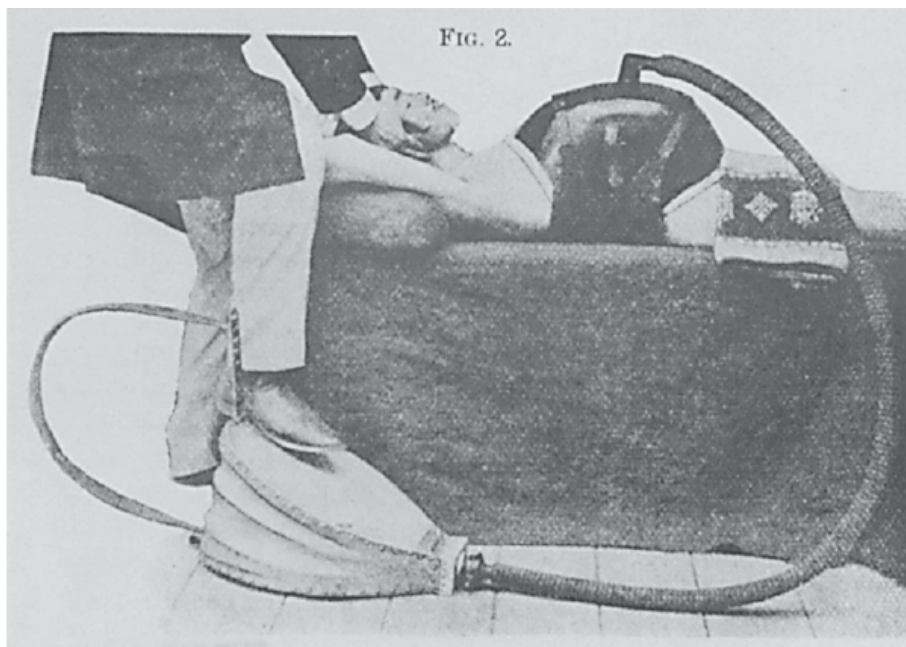
In Egon Braun's resuscitation box, children were seated in a plaster mold with their noses and mouths protruding through a rubber diaphragm. The operator blew through the tube on the right first, compressing the chest. Then suction was applied to the tube expanding the chest. (Used, with permission, from Emerson JH. *Evolution of Iron Lungs*. Cambridge, MA: JH Emerson; 1978:fig. 4.)

Figure 1-22



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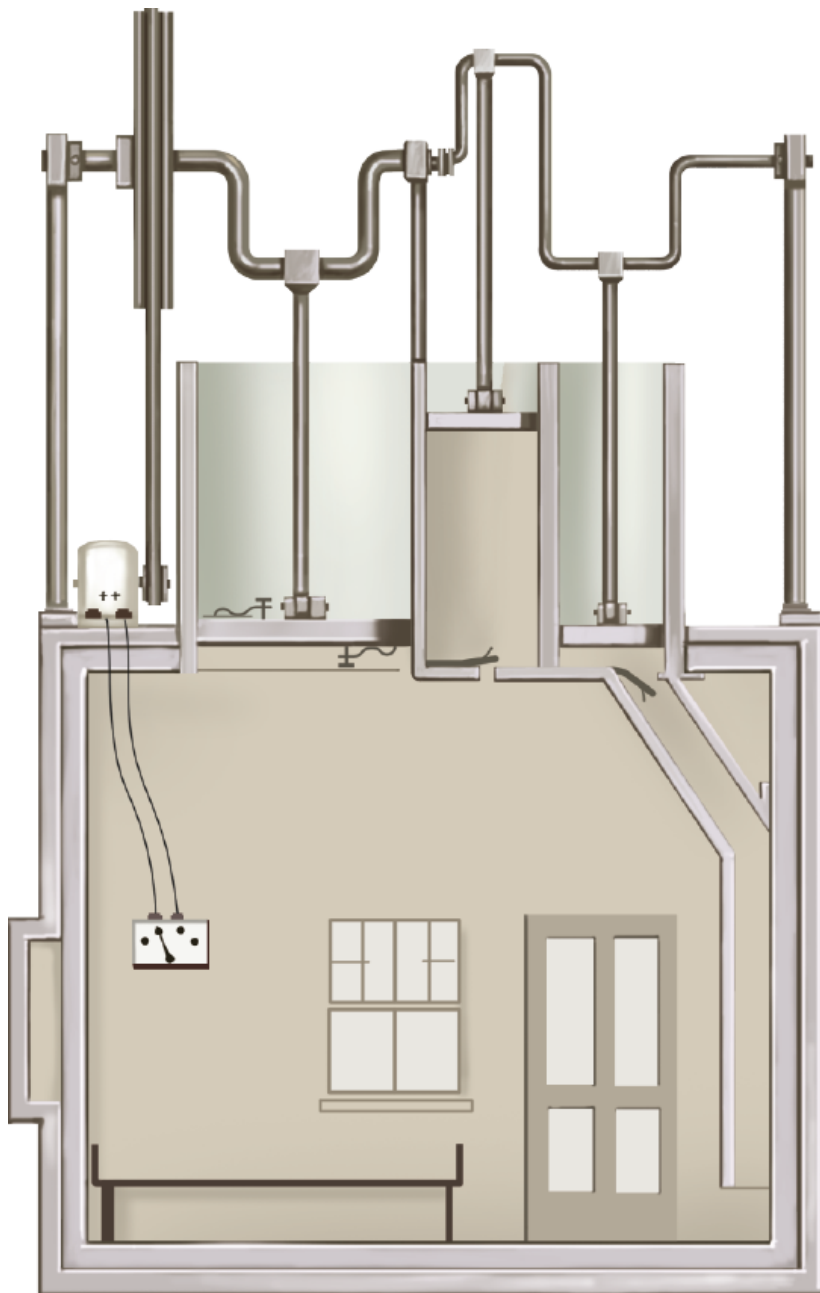


**B**

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**A.** Eisenmenger's earlier version of his cuirass shell. (Used, with permission, from Emerson JH. *Evolution of Iron Lungs*. Cambridge, MA: JH Emerson; 1978:fig. 5.) **B.** The more sophisticated version described in 1904. A foot bellows allows positive and negative pressure to assist expiration and inspiration, respectively. This device is quite similar to chest shells still in use today. (Eisenmenger R. Apparatus for maintaining artificial respiration. *Lancet*. 1904;1:515.)

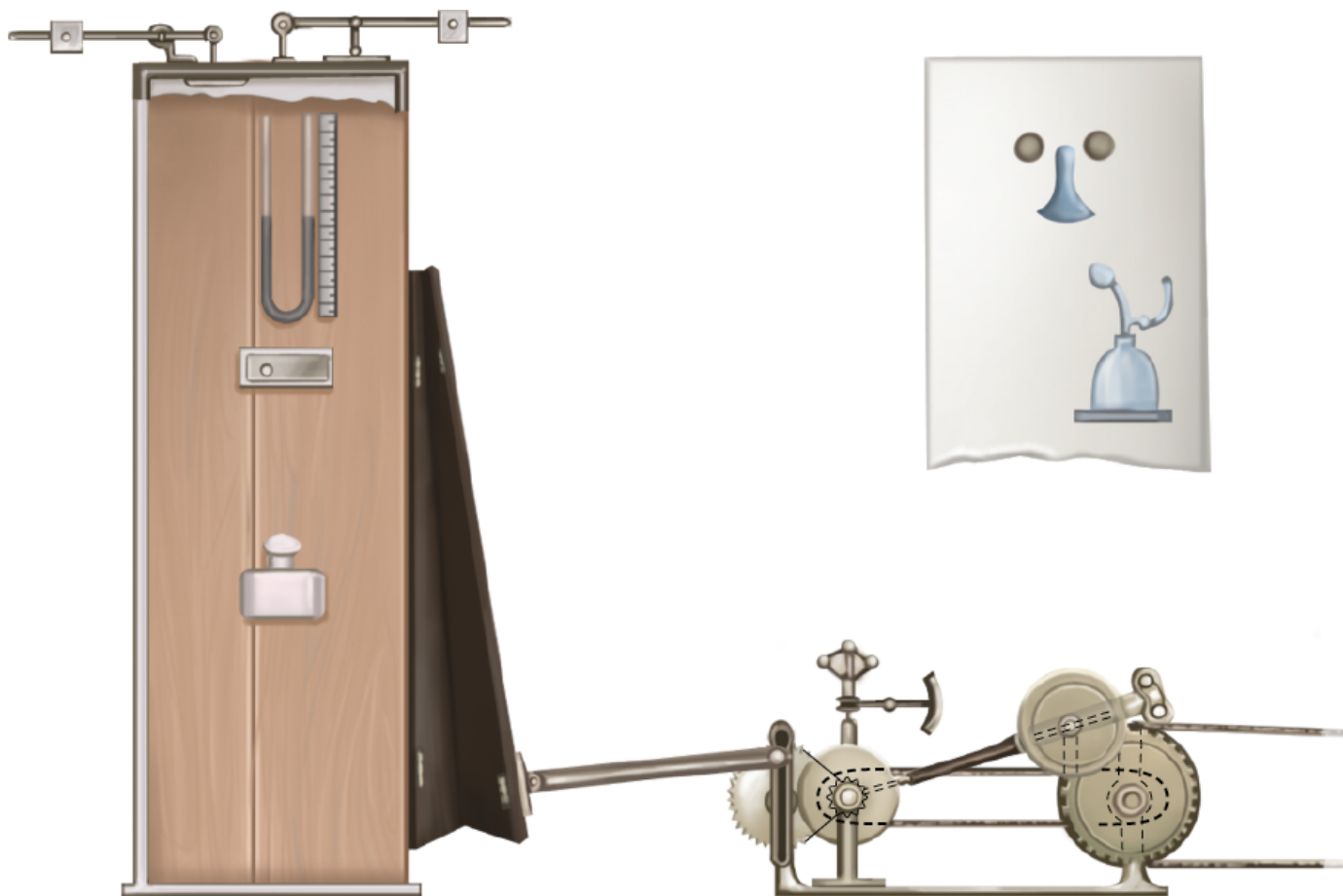
Figure 1-23



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A negative-pressure respirator room, patented by Lord in 1908, provided optimal access to the patient for the nursing staff. (Used, with permission, from Emerson JH. *Evolution of Iron Lungs*. Cambridge, MA: JH Emerson; 1978:fig. 8.)

Figure 1-24

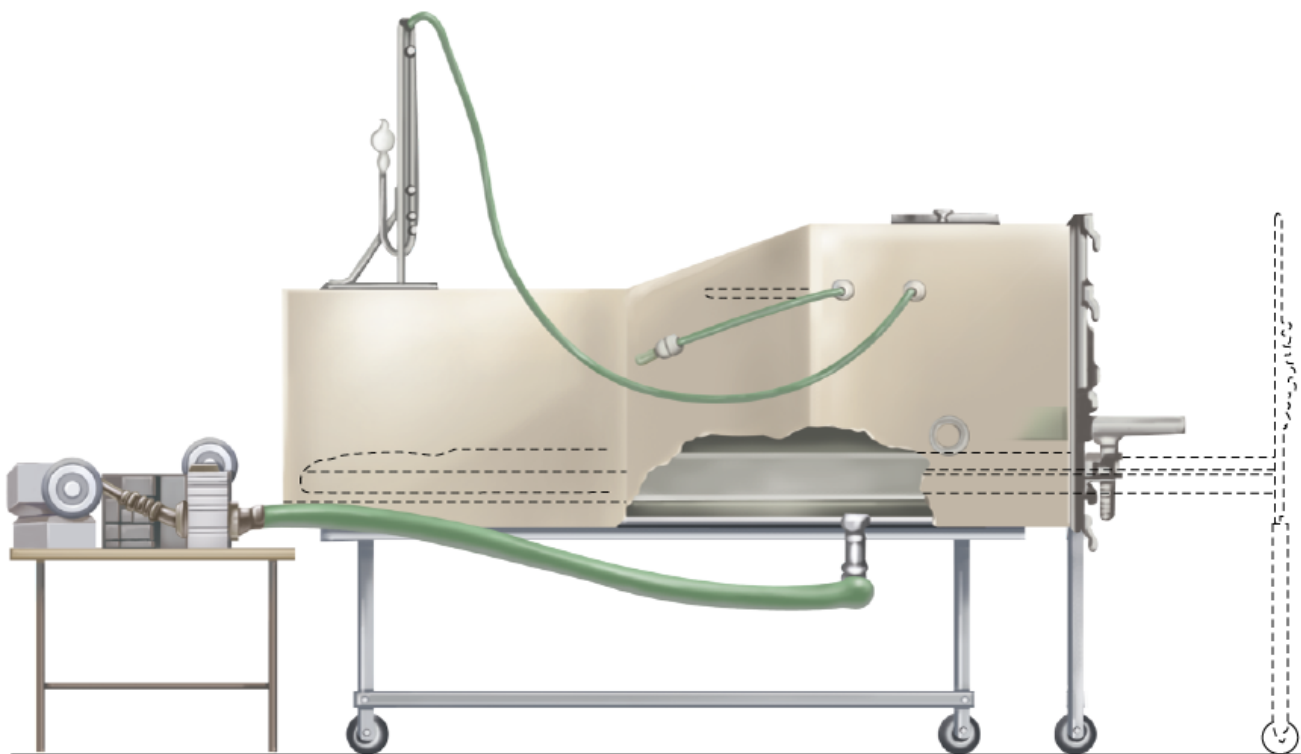


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Severy's negative-pressure ventilator obliged the patient to stand but had a remarkable set of electromagnetic controls and pulleys for adjusting pressure changes within the box. (Used, with permission, from Emerson JH. *Evolution of Iron Lungs*. Cambridge, MA: JH Emerson; 1978: fig. 9.)

The first negative-pressure ventilator to be used successfully in clinical practice on a widespread basis was the Drinker-Shaw "iron lung" developed in 1928 (Fig. 1-25).<sup>80,84-86</sup> With this device, the body was enclosed entirely within a cylindrical sheet-metal tank sealed at the lower end. The patient's head protruded out the upper end through a close-fitting rubber collar. Pressure within the chamber could be either increased or decreased by air blowers. This design is remarkably similar to the spirophore first built by Woillez in 1876. Unfortunately, it suffered from several of the same disadvantages, being cumbersome and inconvenient for patient care. Despite these limitations, this iron lung saved many lives during polio epidemics. When the Consolidated Gas Company of New York paid for large numbers of these machines to be built, their use spread quickly worldwide.<sup>80</sup> A severe poliomyelitis epidemic in 1931 prompted Emerson to build a simplified and improved tank respirator (Fig. 1-26). Because of its low cost, ease of operation, and technologic improvements, the Emerson tank respirator became the mainstay in treating patients with respiratory paralysis from polio<sup>80</sup> until reintroduction of positive-pressure ventilation in the 1950s.

Figure 1-25

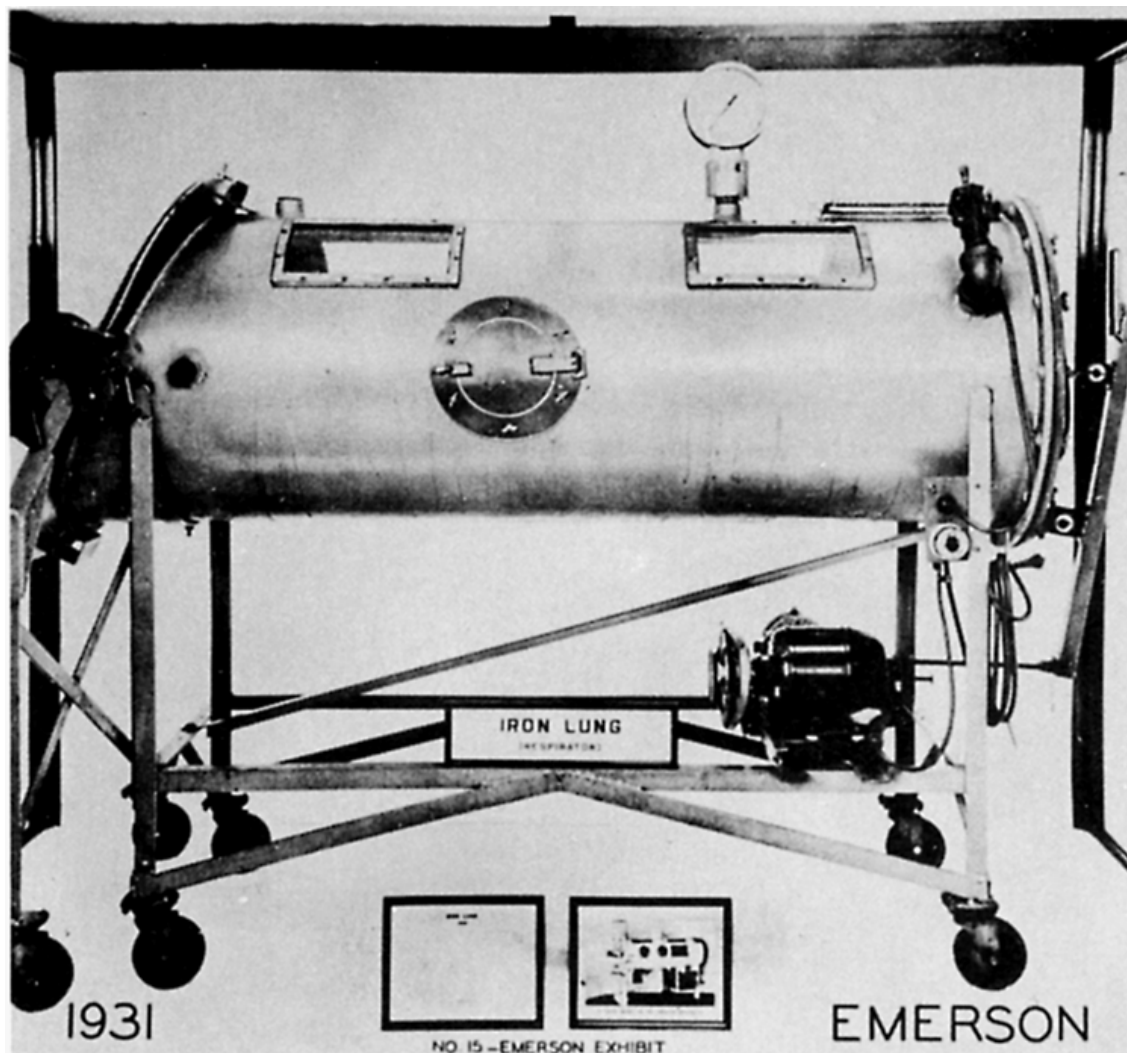


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The Drinker-Shaw iron lung, developed in 1928, had a sliding bed and a close-fitting rubber collar for sealing the patient's neck. The patient's head protruded from the device at the right and rested on a flat support. (Used, with permission, from Emerson JH. *Evolution of Iron Lungs*. Cambridge, MA: JH Emerson; 1978:fig. 13.)

Figure 1-26



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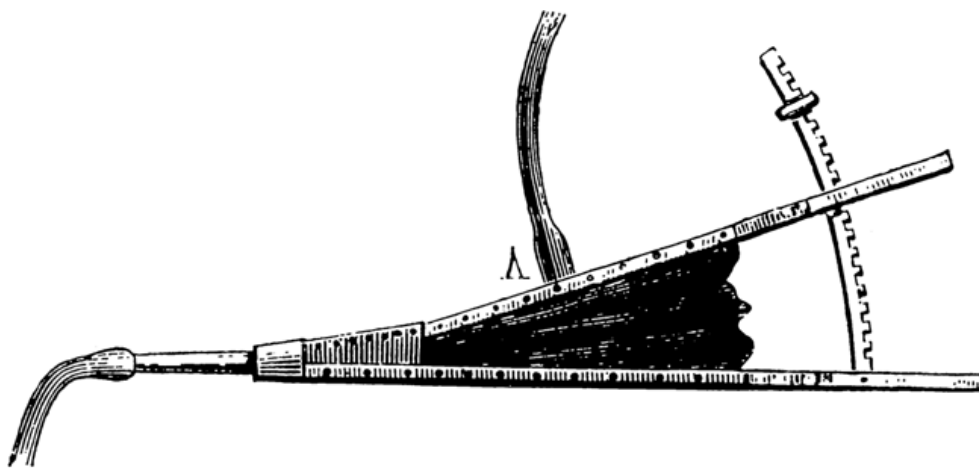
The Emerson tank respirator, built in 1931, used a bellows device to change pressure (hand pumps also were available in case of electricity failure), was less cumbersome and expensive than the Drinker-Shaw iron lung, and was easily opened and closed for nursing care. (Used, with permission, from Emerson JH. *Evolution of Iron Lungs*. Cambridge, MA: JH Emerson; 1978:fig. 15.)

## Positive-Pressure Ventilation

### In the Physiology Laboratory

Although positive-pressure ventilation was disavowed by clinicians, throughout the middle to late 1800s, physiologists relied increasingly on positive-pressure ventilation in animal experiments. Hering and Breuer used this technique to examine how alterations in lung volume influenced the vagi in 1868.<sup>87</sup> Bert, in his studies on blood oxygen and carbon dioxide content, wrote of giving animals sufficient curare to induce total paralysis and of providing artificial ventilation through a tracheostomy tube in 1878. A bellows with a graduated handle for controlling tidal volume proved to be quite effective in these experiments (Fig. 1-27).<sup>15</sup> Pflüger<sup>26</sup> and Head<sup>88</sup> described complex experiments in which cuffed tubes were used to ventilate isolated portions of the lung. Bowditch wrote of a simple but reliable volume-cycled ventilator used for animal studies as a standard piece of laboratory equipment at Harvard in 1879 (Fig. 1-28).<sup>89</sup>

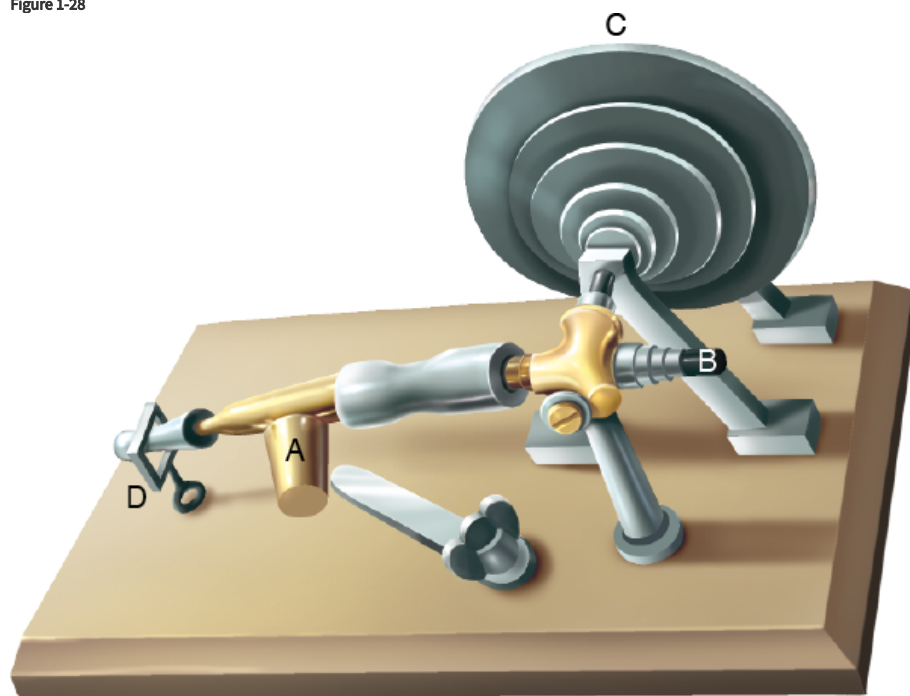
Figure 1-27



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A bellows with graduated handle for controlling tidal volume was used by Bert to ventilate paralyzed animals in 1878. (Used, with permission, from Bert.<sup>15</sup>)

Figure 1-28



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The volume-cycled ventilator used in the Harvard Physiology Department in 1870s. Air from a bellows or tromp enters A and passes to the animal through B. The respiratory rate is determined by adjusting the driving band on cone C. The amount of air entering the animal is determined by screw clamp D, which permits air to escape. (Used, with permission, from Bowditch HP. Physiological apparatus in use at the Harvard Medical School. *J Physiol.* 1879–1880;2:202.)

### In the Operating Room

The reintroduction of positive-pressure ventilation into clinical medicine occurred in two stages. Initially, positive-pressure ventilation was used in the operating room around the turn of the twentieth century. There was a clear and pressing need for positive-pressure ventilation to facilitate thoracic surgery. From the time of Galen, it was appreciated that opening the thorax invariably caused fatal pneumothorax. Vesalius had shown that positive-pressure ventilation could keep an animal's lungs inflated and the animal alive during these operations, but this lesson seems to have been forgotten. Consequently, lung surgery in the nineteenth century was limited to rare cases of draining lung abscesses and bronchiectatic or tuberculous cavities. Although pneumonectomy had been performed successfully in animals by 1881,<sup>90,91</sup> between 1880 and 1920 the mortality rate for thoracic surgery remained high, and these procedures were performed infrequently.<sup>90,92</sup> By 1896, Quénu and

Longuet realized that to have success in thoracic operations, one had “to maintain a difference in pressure between the intra-alveolar air and the surrounding air.” The surgeon could choose either to “lower the extra thoracic pressure, the intrapulmonary tension remaining the same, making it necessary...to operate in a relative vacuum, or to increase the intrabronchial pressure.”<sup>93</sup> Only after positive-pressure ventilation had become a well-established technique during surgery was it applied to nonoperative patients, beginning in the 1950s.

## Tracheal Intubation

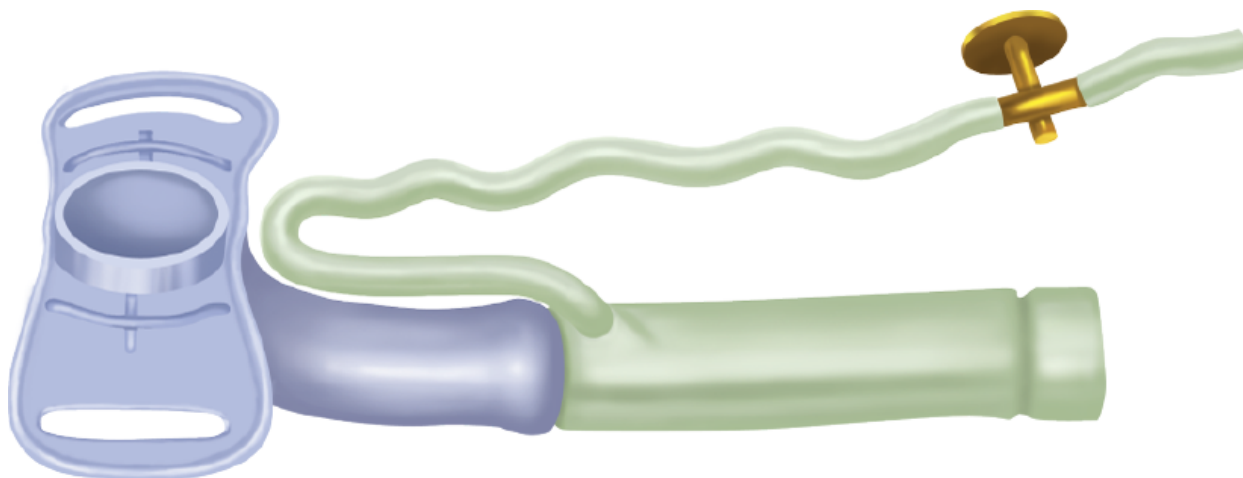
The obvious way to increase intrabronchial pressure would be to follow Vesalius’s example by placing a tube in the trachea and inflating the lungs with a bellows. Positive-pressure ventilation had been well standardized in the physiology laboratory. Although techniques to cannulate the trachea had been developed in the late eighteenth century, translaryngeal intubation still was viewed skeptically by many physicians until the early 1900s.<sup>94</sup> Physicians only became reassured about the usefulness of translaryngeal intubation through work on controlling the airway. The most compelling reason for airway control always had been upper-airway obstruction. Tracheostomy historically was a well-known method of gaining access to the airway in such situations. Indirect references to this technique can be found in such ancient texts as the *Rig Veda*, written between 2000 and 1000 BC, and *Eber’s Papyrus*, from about 1550 BC. Alexander the Great reputedly performed a tracheostomy with his sword in 400 BC on a soldier who was choking on a bone.<sup>95</sup> According to Frost<sup>96</sup> and McClelland,<sup>97</sup> Asclepiades of Bithynia was the first surgeon to perform tracheostomy routinely, around 100 BC. The Roman Antyllos also was noted for his skill in this procedure in 340 AD.<sup>96</sup> Although few surgeons were reported to have performed this procedure during the Dark Ages, Brasavola reintroduced the technique to the medical community in 1546. In 1833, Trousseau clearly demonstrated the lifesaving value of tracheostomy for managing upper-airway obstruction with his report on 200 of these operations for diphtheria.<sup>96,97</sup>

A small number of pioneering physicians experimented with techniques and devices to cannulate the trachea through either the nose or the mouth (translaryngeal intubation) in the late eighteenth century as a practical alternative to tracheostomy for managing upper-airway obstruction. Depaul, who succeeded Chaussier at the maternity hospital in Paris, successfully modified Chaussier’s tubes for managing neonatal respiratory distress.<sup>73</sup> Bouchut used a silver truncated cone, “a little smaller than a thimble,” to dilate the larynx in two children with diphtheria in 1858.<sup>98</sup> Although transiently successful, Trousseau discouraged this approach and confirmed tracheotomy as the preferred approach. MacEwen of Glasgow used translaryngeal intubation to manage a small number of patients with upper airway obstruction and probably diphtheria in 1880.<sup>99</sup> O’Dwyer of New York proved to be the most persistent advocate of intralaryngeal tubes for managing upper airway obstruction. He was appointed to the medical staff of a foundling home in 1872. At that time the leading cause of infant mortality was diphtheria of the larynx. Even with tracheotomy, the survival rates for this condition were abysmal. O’Dwyer worked from 1880 to 1885 on developing a device for intubating the larynx, but his initial experience with a bivalved device proved unsuccessful. Despite initial skepticism by the medical community about this technique, O’Dwyer persevered and later developed a more successful, simple, open-tube approach.<sup>100–102</sup> O’Dwyer, through his efforts over years and experience in hundreds of patients, should be recognized as the first physician to effectively advocate the use of translaryngeal intubation for managing upper-airway obstruction.<sup>98</sup>

## Tracheal Anesthesia

As clinicians began to appreciate the value of translaryngeal intubation for managing upper-airway obstruction, anesthesiologists slowly began to grasp how useful this technique might be for administering anesthesia in the operating room. Ether anesthesia had first been used during an operation by Morton in 1846.<sup>103</sup> Snow, in 1858, gave rabbits chloroform vapor through a tracheostomy tube.<sup>104</sup> Trendelenburg apparently adapted this method for use in patients during operations on the mouth and larynx. A practical problem limiting these operations at the time was aspiration of blood into the lungs during the procedure. Trendelenburg solved this problem by devising a cuffed tracheostomy tube for sealing the airway in 1869 (Fig. 1-29).<sup>73</sup> With this cuffed tracheostomy tube in place, inhalational anesthesia could only be given practically through the tube itself. MacEwen’s original work on translaryngeal intubation for upper-airway obstruction included one case in which successful anesthesia given through the tube allowed surgical removal of a pharyngeal tumor.<sup>98</sup> Maydl of Prague and Eisenmenger of Vienna both described cases of upper-airway surgery in 1893 using endotracheal anesthesia.<sup>105</sup>

Figure 1-29



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Trendelenburg first used this type of tracheostomy tube with inflatable cuff to prevent aspiration during operations on the mouth and larynx in 1869. (Used, with permission, from Lomholt N. A new tracheostomy tube: I. cuff with controlled pressure on the tracheal mucus membrane. *Acta Anaesthesiol Scand*. 1967;11(3):312.)

Tuffier and Hallion, in 1896, ventilated animal lungs through a translaryngeal tube to prevent collapse during surgery.<sup>106</sup> Doyen improved their techniques.<sup>107</sup> Fell was well aware of the use of forced respiration, using a bellows to administer positive-pressure ventilation through a tracheotomy tube, from animal experiments, but had never heard hint of this technique possibly being useful in humans. He realized, though, that forced respiration might be a practical method for managing patients with respiratory paralysis from opiate overdose.<sup>108</sup> Fell considered that “if the respirations could be kept up by suitable means for a sufficient time to permit the elimination of the poison, life might be saved. He reported use of this technique in both opiate and anesthetic drug overdose.<sup>109</sup> O’Dwyer modified Fell’s device so that it could be attached to a translaryngeal tube, and soon the Fell-O’Dwyer apparatus for mechanical ventilation was marketed. Matas realized that if anesthesia could be administered readily through the apparatus, the Fell-O’Dwyer ventilator would be well suited for managing intrathoracic surgery. Matas modified this ventilator successfully and indicated that the new machine indeed was effective (Fig. 1-30).<sup>107,110</sup>

Figure 1-30



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The Fell-O' Dwyer apparatus with modifications by Matas for delivering anaesthesia. (Used, with permission, from Matas R. Intralaryngeal insufflation. *JAMA*. 1900;34:1468–1473.)

The interest in translaryngeal intubation for administering anesthesia and facilitating positive-pressure ventilation was stuttering at first. Kuhn apparently used this technique regularly and with great success. He experimented extensively with both the size and position of the tracheal tube, even using separate tubes for inhalation and exhalation.<sup>111</sup> In 1909, Meltzer and Auer modified this technique. Instead of using a tracheal tube that nearly approximated the diameter of the trachea and allowing the patients to inhale and exhale through that same tube, these physiologists used a narrow-bore tube. Air was blown into the lung through the tube and allowed to escape from the lung between the external wall of the tube and the trachea. This technique was referred to as *endotracheal insufflation*.<sup>112</sup> Ellsberg was the first to use endotracheal insufflation on a patient,<sup>113</sup> but others criticized the technique. Meyer disparagingly called Meltzer's insufflation method the "blowpipe apparatus." Meyer cited numerous concerns (e.g., risks of aspiration through the tube; incomplete control of airway pressure, especially when closing the chest; unreliable administration of anesthesia; and difficulties placing the tracheal tube correctly) and concluded that endotracheal insufflation was not suitable for thoracic surgery.<sup>94</sup>

## Differential Pressure

Around 1904, Sauerbruch devised a working model of a cabinet that generated negative pressure around the lung. Sauerbruch built a small airtight operating room with the patient's body and the surgeon inside. The patient's head extended out of the room. By applying suction to this room, differential pressure was created across the pleural surface, atmospheric pressure within the bronchial tree, and negative pressure outside the lung (Fig. 1-31).<sup>64</sup> Meyer built a much larger version of a negative-pressure apparatus. The entire operating room was subjected to suction, and a small chamber was built within the operating room to enclose the patient's head and the anesthetist. This chamber either could be kept at atmospheric pressure or could be subjected to positive pressure (Fig. 1-32).<sup>114</sup>

Figure 1-31

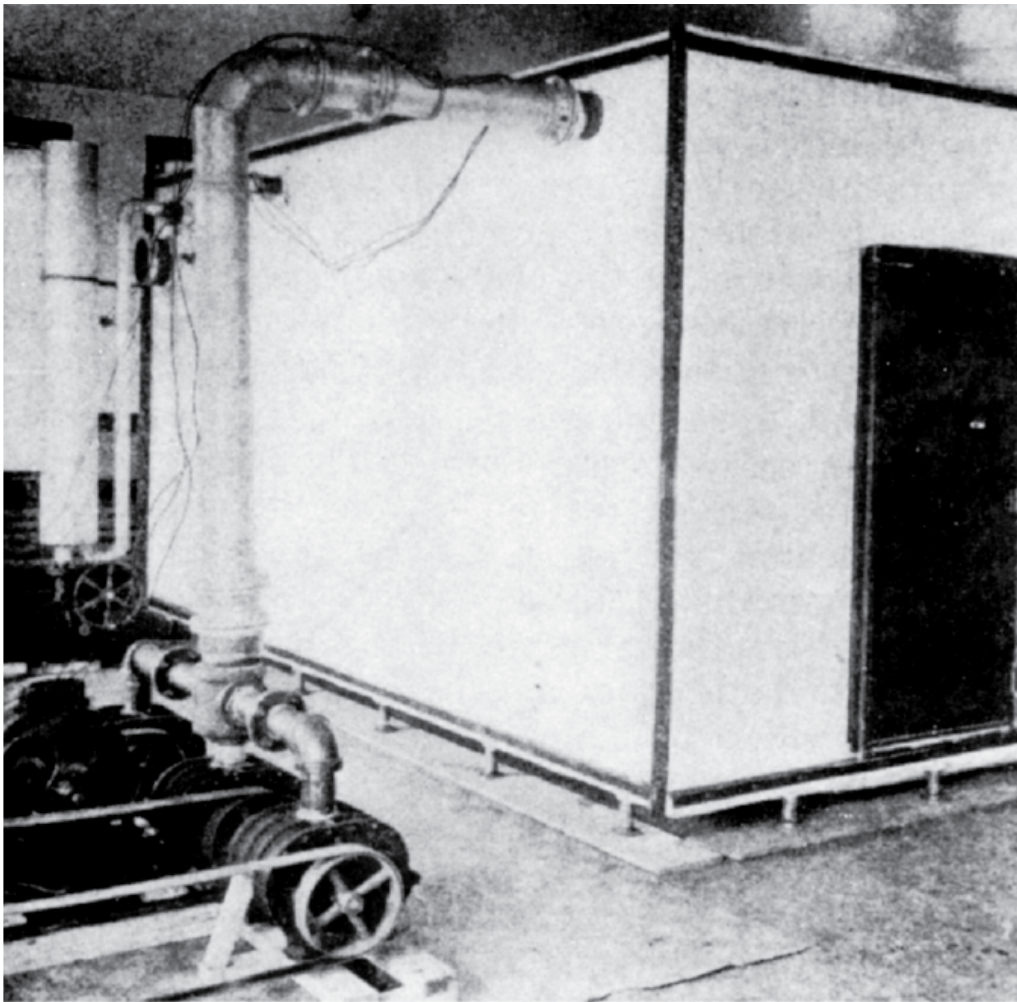


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The differential-pressure cabinet developed by Sauerbruch placed the surgeon inside a small chamber. Suction was applied to this chamber. With the animal's (in this case) or patient's head outside the chamber, the intrabronchial pressure remained at effective sea level. This differential pressure maintained lung inflation when the thorax was open. (Used, with permission, from Morsch.<sup>64</sup>)

Figure 1-32



**A**

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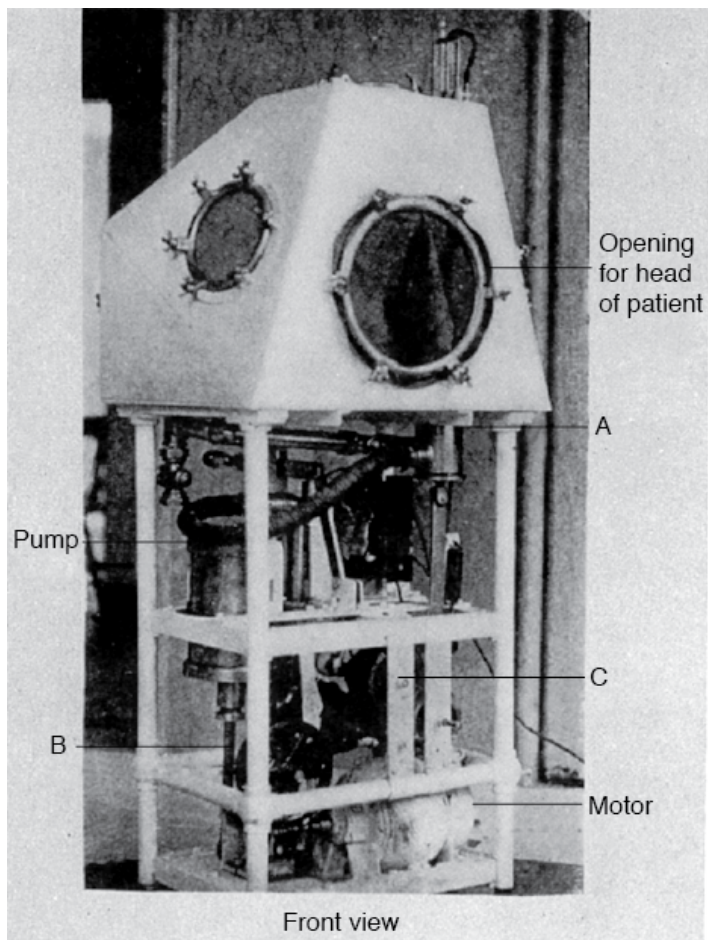
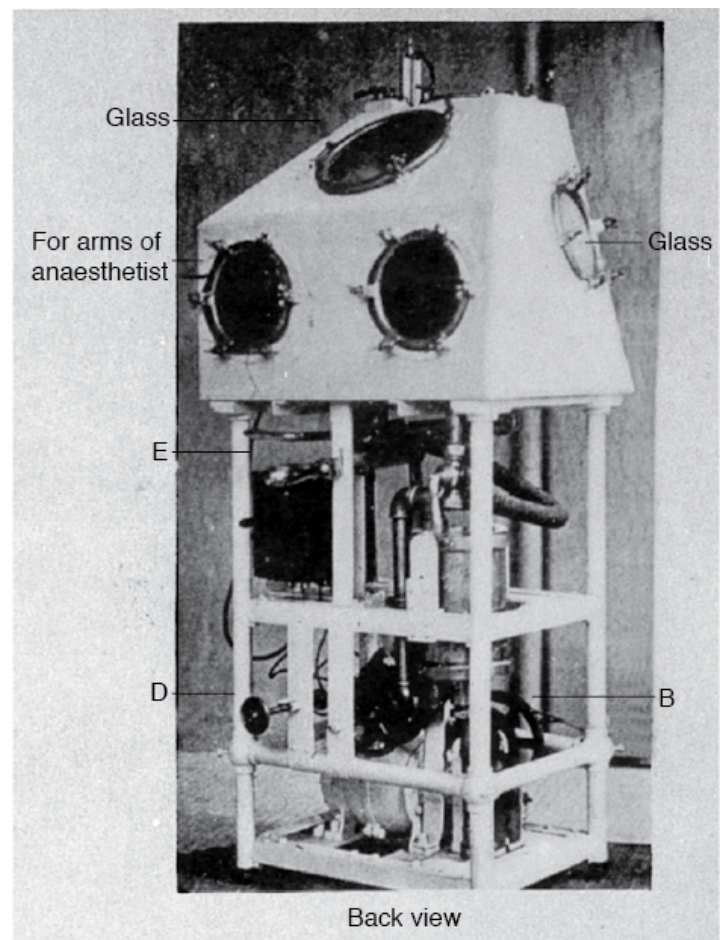
B

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A larger version of the differential-pressure cabinet was constructed by Meyer. **A**. The large chamber, shown from an outside view on the left, has suction applied to it. Inside this chamber, the patient is placed on a table with the patient's head inserted into a smaller chamber. Within this smaller chamber (**B**), kept at positive pressure, resides the anesthetist. (Used, with permission, from Meyer W. Pneumonectomy with the aid of differential air pressure. *JAMA*. 1909;77:1984.)

Surgeons also had the option of using devices that employed positive pressure without intubation to inflate the lungs. Brauer is credited as devising the first positive-pressure cabinet in 1904. This apparatus was large enough for the patient's head to fit inside. Positive pressure in the cabinet would be transmitted to the lungs during the patient's respiratory efforts.<sup>64,115–117</sup> Modifications of the positive-pressure cabinet were put forth by Murphy,<sup>115</sup> Green,<sup>116</sup> and Green and Janeway (Fig. 1-33).<sup>117</sup> As an alternative to cabinets, other surgeons used face masks and helmets to supply positive-pressure ventilation (Fig. 1-34).<sup>118–120</sup>

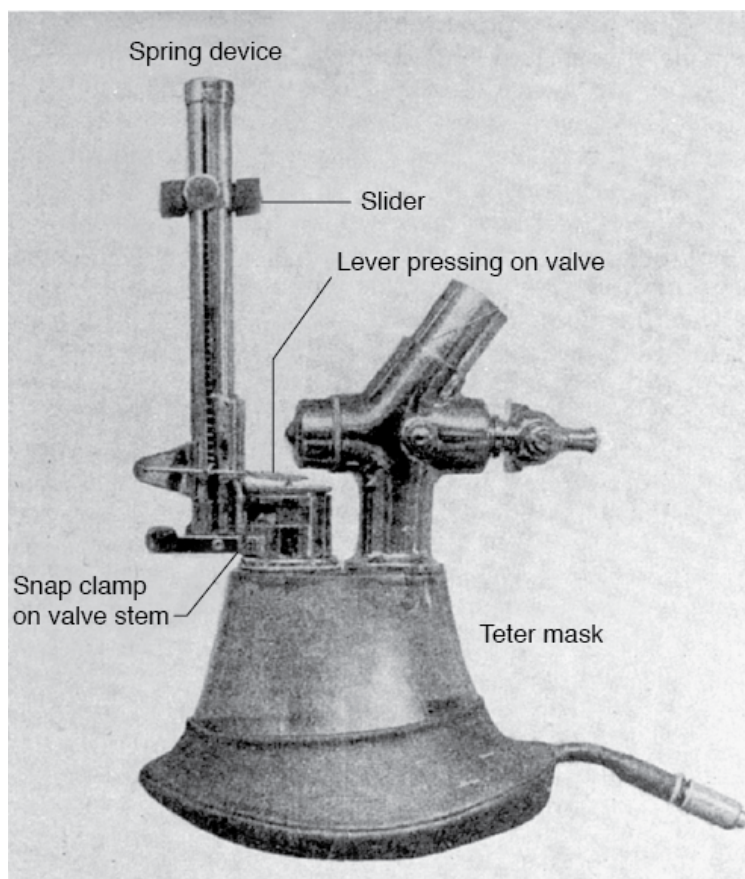
Figure 1-33

**A****B**

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Green and Janeway proposed this positive-pressure chamber in 1910. The patient's head is inserted into the chamber (A). The anesthetist's arms also can reach into the chamber (B). (Used, with permission, from Green NW, Janeway HH. Artificial respiration and intrathoracic esophageal surgery. *Ann Surg.* 1910;52:58-66.)

Figure 1-34

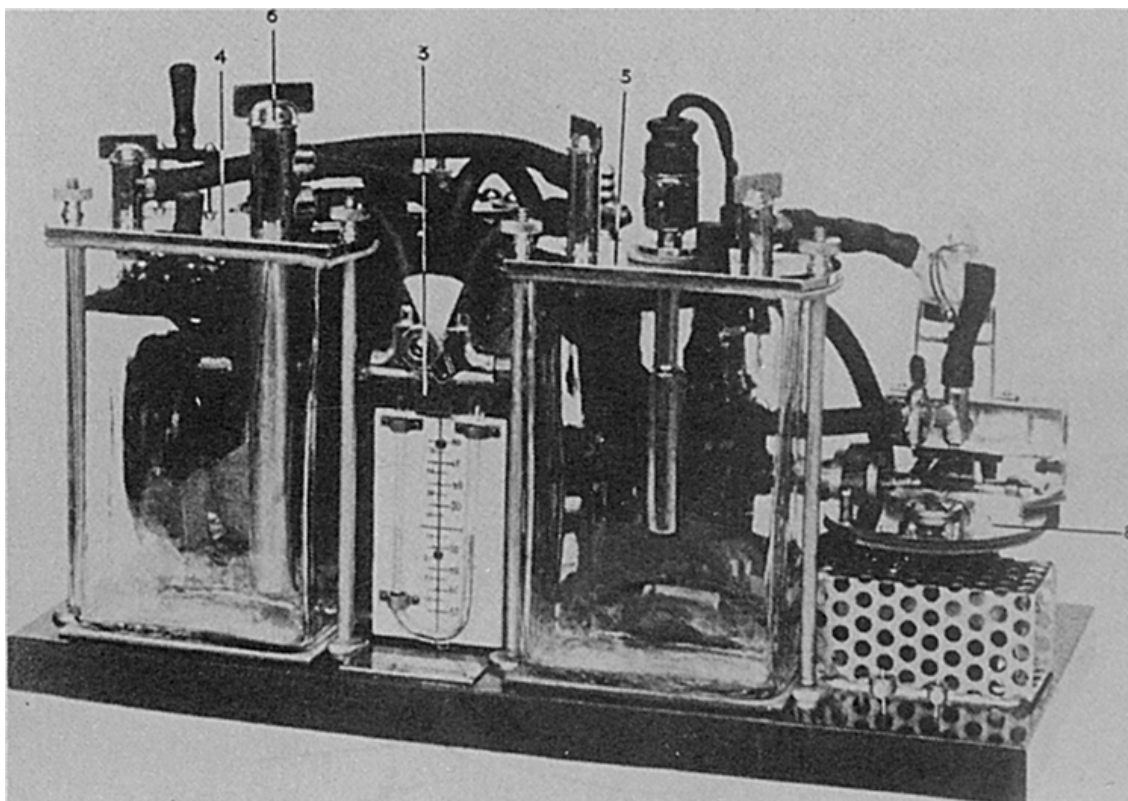


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A positive-pressure mask used by Bunnell in 1912. (Used, with permission, from Bunnell S. The use of nitrous oxide and oxygen to maintain anesthesia and positive pressure for thoracic surgery. *JAMA*. 1912;58:836.)

Several factors favored the endotracheal insufflation method for both anesthesia and lung inflation during thoracic operations. The differential-pressure chambers were either large and cumbersome or small and confining, and all were expensive. A device for endotracheal insufflation, which was portable and easy to use, was built in 1910 by Elsberg for less than \$100,<sup>121</sup> although more complex devices also were built (Fig. 1-35). The positive-pressure cabinets limited access to the patient's head during the procedure. Positive-pressure masks were not reliable in maintaining lung inflation. Techniques and devices for facilitating translaryngeal intubation were being developed quickly. Dorrance<sup>122</sup> described a simple endotracheal tube made of flexible rubber with an inflatable cuff at its distal end. Janeway developed the first modern laryngoscope, an invaluable aid for translaryngeal intubation.<sup>64</sup> Jackson was instrumental in providing clear guidelines for performing translaryngeal intubation.<sup>123</sup> With these developments, endotracheal insufflation became firmly established, and new devices for anesthesia and ventilation by insufflation were developed quickly.<sup>124,125</sup> Many other devices for positive-pressure ventilation, including the "pulmotor" portable resuscitator, were devised in the early 1900s.<sup>64</sup>

Figure 1-35



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A sophisticated and complex ventilator built by Janeway in 1912. (Used, with permission, from Janeway HH. An apparatus for the intratracheal insufflation. *Ann Surg.* 1912;56:328–330.)

## Translaryngeal Intubation

By World War I, endotracheal intubation had become an invaluable method for enabling extensive plastic facial reconstructions. However, anesthetists began to express dissatisfaction with the insufflation technique for anesthesia and ventilation. Insufflation did not protect from aspiration, especially during upper-airway operations. In these procedures, pharyngeal packing to prevent aspiration required placement of two tubes, one for insufflation and the other for exhalation. Placement of two tubes was technically difficult. Anesthetists would much prefer to use a cuffed tracheal tube, which, of course, was not feasible with insufflation. Anesthetists were finding that nitrous oxide was a better anesthetic agent than chloroform or ether, but it was very expensive to administer by insufflation. Periodic deflations of the lung usually were required, with insufflation to ensure adequate carbon dioxide removal. As a consequence of these problems with insufflation, Magill and Rowbotham returned to Matas's old "inhalation" method. They used a tracheal tube large enough to allow both inhalation and exhalation. A balloon cuff could be attached to the outer distal end of the tube to prevent aspiration.<sup>105,126,127</sup> Rowbotham also was instrumental in popularizing nasotracheal intubation.<sup>128</sup> A number of cleverly designed machines were produced before World War II that could be used to administer anesthesia and artificial ventilation via positive-pressure rhythmic "insufflation" (ventilation) through these large-diameter tubes.<sup>64,129,130</sup> By 1934, Guedel and Treweek described apneic anesthesia, or purposely giving enough anesthesia to cause complete respiratory paralysis. Artificial respiration by rhythmic bag ventilation adequately supported the patient during apnea. This technique provided the "quiet" field necessary for abdominal and thoracic surgery.<sup>131</sup>

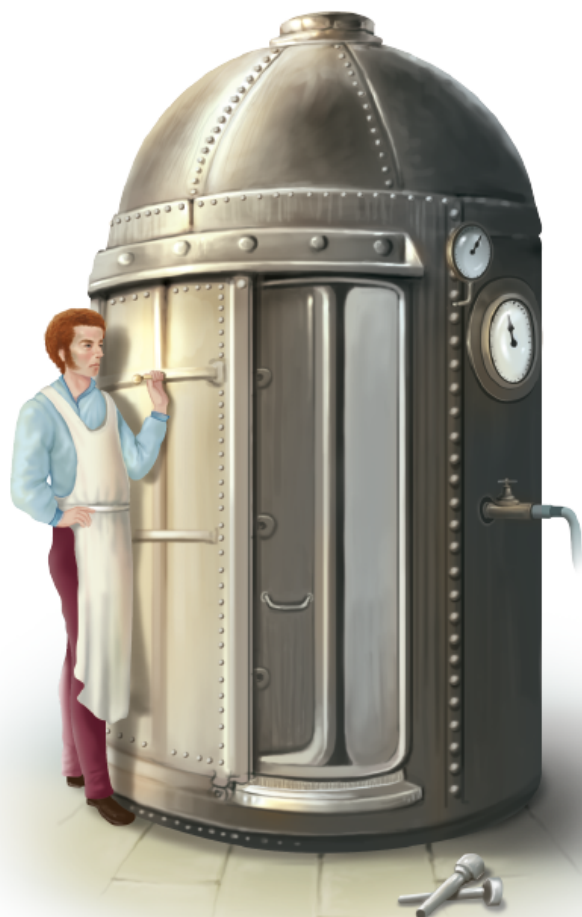
Further favoring the use of translaryngeal intubation was recognition of retained pulmonary secretions as a cause of postoperative morbidity and mortality. As pointed out by Jackson in 1911, "when tracheal and bronchial secretions are in excess of the amount required properly to moisten the inspired air, they become a menace to life unless removed."<sup>132</sup> Postoperative atelectasis, attributed to retention of thick bronchial secretions and inhibition of coughing, was first described in 1928.<sup>133</sup> Bronchoscopy and intratracheal suctioning were the earliest techniques advised for removing secretions,<sup>132,134,135</sup> but it became apparent that easy access to the tracheobronchial tree for repeated suctioning might be necessary in difficult cases. Tracheostomy was recommended as early as 1932 specifically for this problem in polio patients,<sup>136,137</sup> and later for patients with a variety of surgical problems.<sup>138–141</sup> Physicians soon realized that suctioning through a large tube placed translaryngeally might be just as effective as through a tracheotomy tube. In the 1940s, case reports began to appear describing the successful and prolonged use of

translaryngeal intubation for tracheobronchial toilet.<sup>142–144</sup> Modification of tracheal tubes to include “Murphy eyes” at the tips probably has reduced the likelihood that these tubes would become occluded by mucus.<sup>145</sup>

## For the Nonoperative Patient

The second stage of the reintroduction of positive-pressure ventilation into clinical practice involved nonoperative patients and occurred dramatically in the 1950s. There had been isolated reports of physicians using positive-pressure ventilation for purely mechanical problems before this time. Bert describes a positive-pressure chamber built by Jourdanet in the 1870s (Fig. 1-36) and used for a variety of mechanical problems.<sup>15</sup> Williams wrote of treating pulmonary disease with a pneumatic differentiation chamber in 1885. The patients were placed within a cabinet, and the air in this cabinet was exhausted by suction. Simultaneously, “antiseptic air charged with remedial agents” was administered to the patient’s mouth. The reduced pressure around the thorax and atmospheric pressure applied to the lungs were thought to dilate the lungs beneficially. Remarkable improvements were described for a wide variety of lung disorders with this device.<sup>146</sup> Fell used a bellows ventilator to manage respiratory depression secondary to opiate overdose in the late 1880s.<sup>108</sup> A remarkable series of studies described the use of positive-pressure respiration for the treatment of pulmonary edema.<sup>147–149</sup> The emphasis of these studies was not to assist respiration—forced respiration by intubation was felt to be unjustifiable<sup>147</sup>—but to use positive pressure to counterbalance the backward pressure on the pulmonary capillaries.<sup>149</sup> The widespread use of positive-pressure ventilation did not begin, however, until its value was demonstrated dramatically during a polio epidemic in Copenhagen in 1952.

Figure 1-36



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Jourdanet used this positive-pressure chamber to treat patients with a wide variety of disorders in the 1870s. (Used, with permission, from Bert.<sup>15</sup>)

## Paralytic Polio

A series of polio epidemics had swept across Europe and the United States in the 1930s and 1940s. Respiratory paralysis secondary to poliomyelitis was an infrequent but feared complication. Even with the best management techniques using iron lungs and cuirass ventilators (Fig. 1-37), the mortality rate for polio-induced respiratory paralysis probably was approximately 85%.<sup>64</sup> In the late summer of 1952, an epidemic struck Copenhagen. Of the first thirty-one patients admitted to Blegdamshospital, Copenhagen's hospital for communicable diseases, during this epidemic with respiratory paralysis, twenty-seven died within 3 days. Out of desperation, Henry Lassen, the chief physician and epidemiologist, called the freelance anesthetist Bjorn Ibsen for consultative advice. After reviewing the medical records and autopsy results, Ibsen made two startling conclusions. First, he felt that in the fatal cases there was insufficient atelectasis within the lungs to make adequate ventilation impossible. Second, he suggested that the increased blood levels of total CO<sub>2</sub> did not reflect metabolic alkalosis, as was generally believed, but rather acute respiratory acidosis. Ibsen's observations about respiratory acidosis were derived directly from work he had performed measuring exhaled carbon dioxide levels in the operating theater. Ibsen, as the anesthetist, had noted that exhaled carbon dioxide levels fluctuated during the course of surgery and could be compensated by more vigorous bag ventilation. Most importantly, when exhaled carbon dioxide levels increased, the patients in the operating theater had developed clammy skin and high blood pressure, similar signs to those found in the paralytic polio patients just before death. Based on these observations, Ibsen suggested inadequate ventilation as the cause of death and advised tracheostomy to allow the operative techniques of positive-pressure ventilation. Lassen was not convinced; the iron lung and cuirass respirators had reliably provided adequate ventilation in the past. Lassen argued that it was unlikely that positive-pressure ventilation would save paralytic polio patients if the underlying disease process actually included extensive brainstem involvement.<sup>150</sup>

Figure 1-37

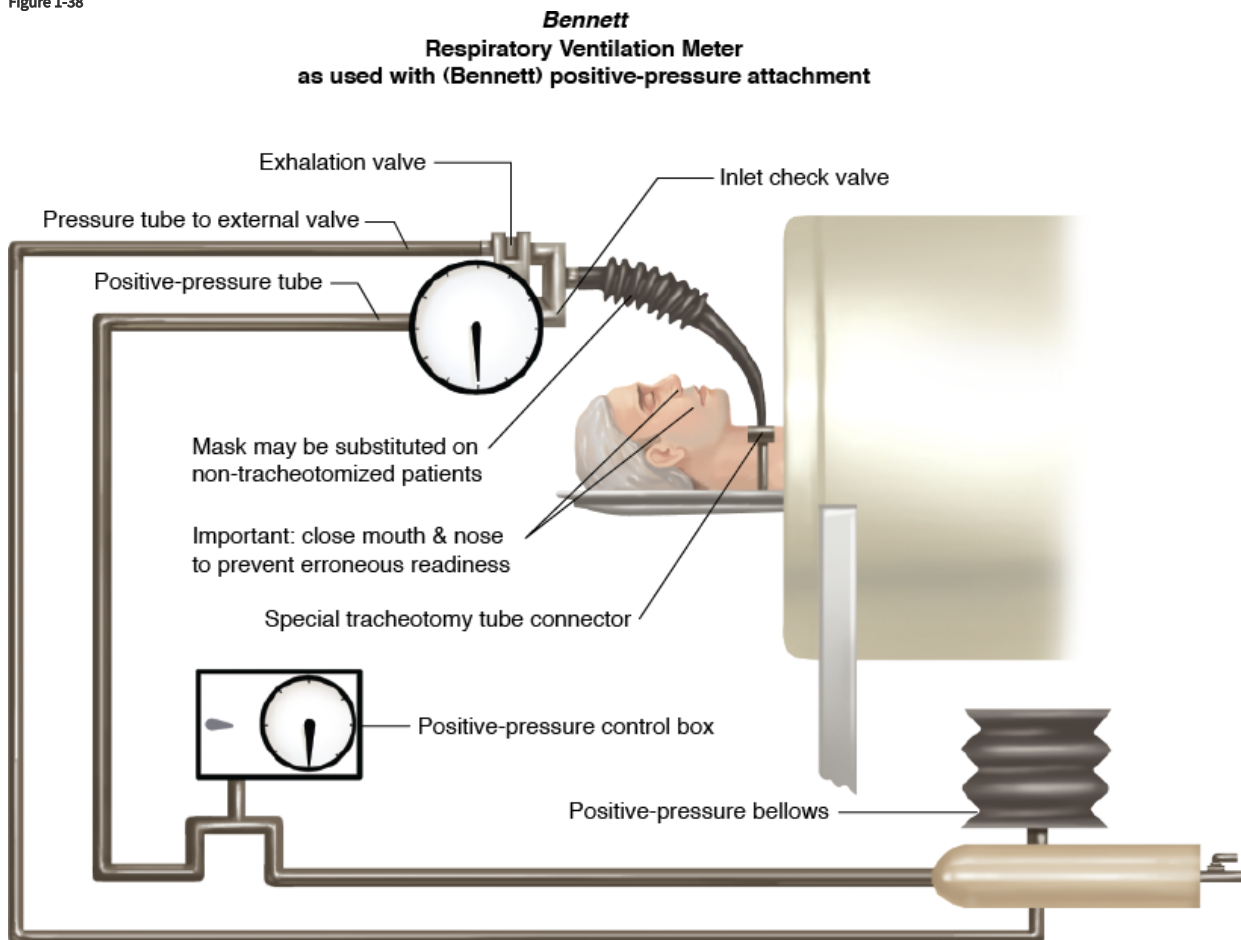


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Young patients with respiratory paralysis from polio being treated in an “iron lung.” (Photograph by Hansel Mieth, courtesy Time Life Pictures, Getty Images, 1938.)

As a counterargument, Ibsen cited recent experience in the United States with a positive-pressure valve capable of providing mechanical positive-pressure ventilation to polio patients. These valves were developed as a result of intense interest by the U.S. Air Force during World War II in using positive pressure to increase altitude tolerance in pilots.<sup>151,152</sup> The unique attribute of these valves, such as the pneumatic balance respirator (PBR), was their ability to convert a continuous positive pressure into intermittent positive pressure. Intermittent positive-pressure breathing was applied in the late 1940s to a variety of medical problems and found to be effective in providing artificial ventilation to an apneic person;<sup>153–155</sup> in managing acute pulmonary edema, acute asthma, and postoperative patients with poor respiratory excursion;<sup>154</sup> possibly in improving oxygenation in various lung diseases;<sup>156</sup> and in administering medications by nebulization.<sup>157</sup> The Bennett valve was adapted as a positive-pressure respirator attachment for the standard tank respirator during the 1948 Los Angeles poliomyelitis epidemic (Fig. 1-38). The PBR supplied intermittent positive-pressure breaths in synchrony with the tank respirator’s inspiratory negative-pressure phase.<sup>158</sup> Use of the PBR along with the tank respirator significantly reduced the case-fatality rate of respiratory paralysis associated with polio.<sup>159</sup>

Figure 1-38



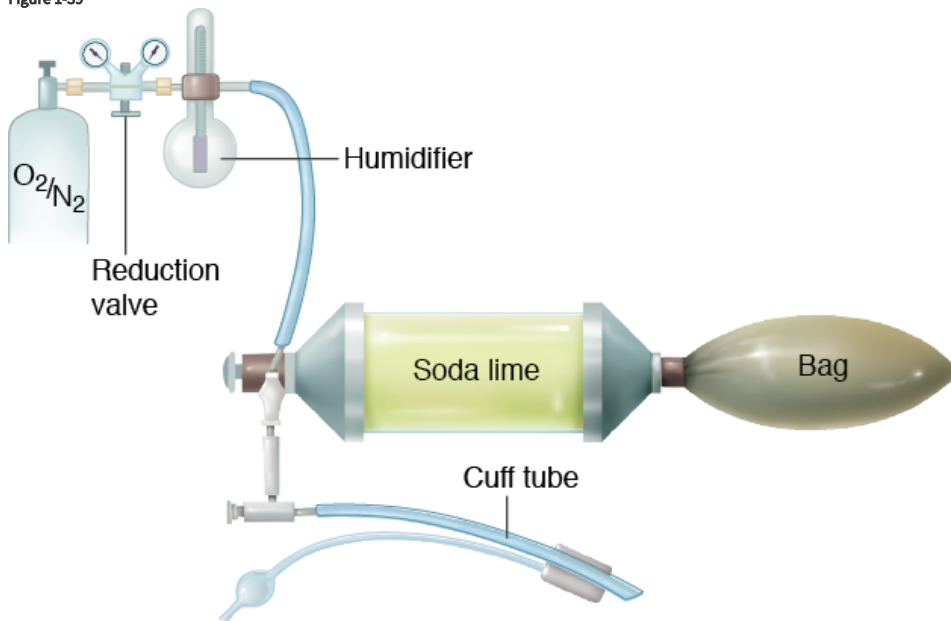
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A schematic of the Bennett positive-pressure valve used via a tracheostomy tube in a patient in an iron lung. (Used, with permission, from Bower AG, Bennett VR, Dillon JB, Axelrod B. Investigation on the care and treatment of poliomyelitis patients. *Ann West Med Surg*. 1950;4:567.)

Lassen eventually agreed to a trial of Ibsen’s theory. The thirty-second patient with respiratory paralysis admitted to Blegdamshospital was the poignant case of a 12-year-old girl. When her condition deteriorated, Ibsen asked a surgeon to perform a tracheostomy, and a cuffed tracheal tube was introduced. During the procedure, the girl became comatose. Ibsen initially was unable to ventilate her effectively. He assumed that retained secretions were the problem, and he suctioned her. Her condition deteriorated further, and many physicians observing the trial began to leave, assuming that the outcome would be fatal. In this desperate situation, Ibsen decided to paralyze the girl. She collapsed immediately, and Ibsen finally was able to ventilate her adequately. Her condition improved immediately.<sup>150</sup> Eventually, arterial blood-gas levels confirmed Ibsen’s suspicions about respiratory acidosis as the cause of death in the previous patients, and positive-pressure ventilation proved successful in substantially reducing the mortality rates from paralytic polio. The only drawback was the equipment available (Fig. 1-39). Only bag ventilation

was possible. During the remainder of the epidemic, it is estimated that 1500 medical and dental students worked around the clock providing bag ventilation by hand to help support these patients.<sup>160,161</sup>

Figure 1-39



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This hand ventilator was used in the Copenhagen polio epidemic of 1952 by hundreds of “ventilators” (i.e., medical students, technicians, volunteers, and others) to save many lives. (Lassen HCA. A preliminary report on the 1952 epidemic of poliomyelitis in Copenhagen with special reference to the treatment of acute respiratory insufficiency. *Lancet*. 1953;1:37–41.)

The Copenhagen experience provided the impetus for a revolution in the medical care of patients with respiratory failure. First, it confirmed the value of positive-pressure ventilation and demonstrated the need for practical mechanical ventilators. Second, by encouraging the grouping of acutely ill patients in certain sections of the hospital and the organization of intensive care for these patients, it led the way for the later development of intensive-care units.<sup>160,161</sup> Third, Ibsen realized that provisions had to be made for resuscitating acutely ill patients in small outlying towns and transporting them to specialized centers.<sup>162</sup> Accordingly, mobile teams were formed with expertise in performing translaryngeal intubation and tracheotomy. After intubation and stabilization, patients could be transferred secondarily. This was obviously the precursor of our present emergency medical system.

### Other Diseases with Inadequate Ventilation

Although the results using positive-pressure ventilation for the respiratory paralytic form of polio were remarkable, application of this technique to other medical problems was slow. Extensive work in pulmonary emphysema and chronic bronchitis had confirmed that in severe cases ventilatory failure was accompanied by high carbon dioxide levels and low oxygen levels. Supplemental oxygen alone seemed to worsen the situation.<sup>163</sup> Sporadic reports described the use of mechanical ventilation for treating this problem. Usually used were body respirators,<sup>163–167</sup> but occasionally either intermittent positive-pressure breathing via a pneumatic balance respirator<sup>164</sup> or hand ventilation<sup>166,167</sup> was used transiently. By 1961, Munck had collected a total of forty-two case reports describing some form of mechanical ventilation for exacerbations of chronic obstructive pulmonary disease (COPD) with successful outcomes in thirty-one.<sup>168</sup> Munck’s group was the first to rely strictly on positive-pressure ventilation through a tracheotomy tube to treat patients with COPD in acute respiratory crises. Their methods emphasized reliance on monitoring arterial blood oxygen, carbon dioxide, and pH levels. The average duration of treatment in their series was 24 days. They emphasized that mechanical ventilation provides a fair chance of “tiding patients with diffuse chronic lung disease over an episode of life-threatening respiratory failure—and of obtaining a reasonable recovery,” provided there is some historical evidence of pulmonary reserve.<sup>168</sup>

Many other groups throughout the 1960s and early 1970s found that positive-pressure ventilation through either a translaryngeal tube or tracheostomy was an effective method of managing acute exacerbations of COPD.<sup>169–178</sup> Conservative treatment, including the use of controlled levels of supplemental oxygen, antibiotics, bronchodilators, and respiratory stimulants, was useful for treating some patients with

acute ventilatory failure complicating chronic lung disease,<sup>179,180</sup> but it was soon recognized that in severe cases with either coma or deteriorating arterial blood-gas values, endotracheal intubation and mechanical ventilation provided the most appropriate alternative.<sup>181</sup>

In 1951, Nilsson recognized the value of translaryngeal intubation for controlling the airway in patients with barbiturate poisoning.<sup>182</sup> He later emphasized that artificial respiration via a mechanical ventilator is essential when barbiturate poisoning causes apnea or respiratory insufficiency.<sup>183</sup> Avoiding the potentially stigmatizing tracheotomy scar also was an important consideration in patients prone to depression. Bjork pioneered the use of positive-pressure respirator treatment in postoperative thoracic surgery patients. Initially, he was conservative in his approach and postponed tracheotomy until the patient was in severe respiratory failure. By the late 1950s, however, he was performing elective tracheotomy after pulmonary resections and cardiovascular surgery and providing “prophylactic” positive-pressure ventilation to prevent atelectasis and to minimize “heavy respiratory work.” He believed that any patient with a small cardiopulmonary reserve, who could become exhausted rapidly following major surgery, would benefit from this approach.<sup>184–187</sup> As these principles were established in thoracic surgery, they were also applied to patients with, for example, crush injury of the chest, pulmonary edema, renal failure, tetanus, pneumonia, or peritonitis. The best results were obtained when respirator treatment was initiated early in the acute illness and not when chances for recovery were nil.<sup>188</sup>

## Modern Respirators

Providing mechanical ventilation on a widespread basis could only be achieved with reliable respirators. Morch built the first clinically proven volume ventilator during World War II in Denmark for use in the operating room. Because of the war, pistons and cylinders for this ventilator were made from discarded sewer pipes.<sup>64</sup> As a direct result of the 1952 polio epidemic in Denmark, Bang constructed a mechanical respirator. Manual ventilation of patients with respiratory paralysis was possible in Copenhagen because of the availability of medical students. In Skive, where Bang practiced, medical students were not available, and Bang’s respirator was a practical necessity. Fortunately, it worked.<sup>189,190</sup> The Engstrom respirator, built for the same reasons, proved to be hugely successful for managing poliomyelitis patients.<sup>191</sup> This volume-cycled respirator also was used by Bjork in his outstanding work in postoperative respiratory care. By 1954, a number of modern automatic respirators had been developed in Europe.<sup>192</sup> Morch was instrumental in bringing the concept of positive-pressure ventilation across the Atlantic to America.<sup>64,193</sup> The incorporation of this technique into standard medical practice apparently was slower in the United States than in Scandinavia.<sup>64</sup> Tank respirators still were used routinely in the United States until the 1960s,<sup>194,195</sup> although the benefits of endotracheal intubation and positive-pressure ventilation were slowly being appreciated.<sup>194,196</sup> Since the 1950s, an enormous number of practical ventilators have been introduced for everyday use. This was paralleled by a significant increase in the number of patients receiving mechanical ventilation in American hospitals throughout the 1960s. Pontoppidan found a fivefold increase in artificial respiration cases at the Massachusetts General Hospital between 1960 and 1968.<sup>197</sup>

## Intensive Care

Organizing the “intensive care” that patients with ventilatory failure required was a substantial logistical problem. The Danes had congregated respiratory patients in special units. By the middle to late 1960s, widespread experience had accumulated with intensive care units specifically designed for managing patients requiring sophisticated respiratory care and mechanical ventilation.<sup>198–202</sup> By the 1970s, several centers reported impressive reductions in mortality rate through reliance on the intensive care unit approach.<sup>203–205</sup>

## Adequacy of Ventilation

Until arterial blood-gas machines became available commercially, physicians had to rely on laborious and exacting methods of measuring blood oxygen and carbon dioxide levels. Astrup and others emphasized that measuring  $\text{PaO}_2$ ,  $\text{PaCO}_2$ , and pH should be the ultimate goal for determining how well the respirator actually was ventilating the patient.<sup>206,207</sup> By 1957, it was realized that intermittent positive-pressure breathing was not always effective in reducing hypercapnia in emphysema. In fact, the more severe the obstructive lung disease, the less effective pressure-cycled respirators seemed to be in producing hyperventilation.<sup>208</sup> Conversely, pressure-cycled respirators easily could overventilate a patient and convert a dangerous acidosis into an equally dangerous alkalosis.<sup>209</sup> The issue of pulmonary encephalopathy secondary to hypercapnia was of serious concern to physicians,<sup>210</sup> as was the realization that mechanical ventilation could be followed by paradoxical central nervous system acidosis.<sup>211,212</sup>

Supplemental oxygen had been used to treat numerous medical ailments since Priestley and Scheele had identified “pure air” in the 1770s. By the beginning of the twentieth century, the physiologic benefits of oxygen therapy were better understood, and physicians became more interested in using oxygen specifically to treat respiratory disorders. The intravenous injection of oxygen was advocated by Tunncliffe and Stebbing in 1916,<sup>213</sup> but attracted little interest. Haldane devised an apparatus for supplying controlled amounts of oxygen<sup>214</sup> that was used on soldiers exposed to suffocating gases during World War I. Because patients with pneumonia had low blood-oxygen levels, Meakins used the Haldane apparatus to treat hypoxic patients with pneumonia.<sup>215</sup> Stadie constructed and used an oxygen chamber for treating pneumonia in 1922.<sup>216,217</sup> Barach administered oxygen through tents, nasal catheters, and mouth funnels in 1926.<sup>218</sup> Physicians believed that oxygen was useful in reducing the mortality rate in pneumonia but did not understand why it was effective.<sup>219,220</sup> The belief that oxygen treatment was beneficial unfortunately led to its indiscriminant use. It was not appreciated that supplemental oxygen given to a patient with COPD who had an acute ventilatory crisis with hypercapnia could result in paradoxical hypoventilation and worsened respiratory acidosis. Campbell played a leading role in recognizing this problem and devising a device, the venturi mask, for administering oxygen in a controlled fashion with a reduced risk of carbon dioxide retention.<sup>221–223</sup>

Just as oxygen use generally was believed to be beneficial for treating respiratory disorders, the use of oxygen was incorporated routinely into many early mechanical ventilators. This approach was justified in some cases because oxygen requirements to achieve adequate arterial oxygenation often were surprisingly high.<sup>224</sup> Mean inspired oxygen levels in the early respirators, however, could not be regulated accurately. Substantial and occasionally dangerous variations in inspired oxygen levels were found when ventilators were compared directly.<sup>225–227</sup> Nash and coworkers pointed out the potential gravity of this problem when they linked, for the first time in humans, diffuse alveolar damage (or the *respirator lung syndrome*) with the prolonged use of ventilators delivering a high inspired oxygen concentration.<sup>228</sup>

As an interesting aside, Frumin and coworkers found that insertion of an expiratory resistance increased arterial oxygen levels in anesthetized, paralyzed, artificially ventilated humans.<sup>229</sup> Manipulation of airway pressure by immersion of the exhalation limb from a tracheotomy tube 1 to 4 cm under water had been shown previously to improve ventilation in patients with multiple rib fractures.<sup>230</sup> These simple measures for increasing airway pressure, termed variously the *positive expiratory pressure plateau*,<sup>231</sup> *continuous positive airway pressure*,<sup>232,233</sup> and later, *positive end-expiratory pressure*, were to prove enormously successful in improving oxygenation in patients with both adult and infant respiratory distress syndromes.

## Quality Control of Ventilators

As the need for ventilators, gas cylinders, connectors, oxygen-delivery masks, and myriad other types of respiratory equipment increased, the number of manufacturers producing this equipment proliferated. Quality control among manufacturers varied and manufacturers produced equipment of various size specifications, making integration of breathing circuits difficult. Anesthesiologists were particularly concerned with these problems because of the difficulties they had encountered in establishing universal color codes for anesthetic gas cylinders during World War II. Carbon dioxide cylinders in Britain were painted green, but U.S. oxygen cylinders also were painted green. Inevitably, when U.S. and British anesthesiologists worked together, cylinders were filled with the wrong gas and deaths occurred.<sup>234–236</sup>

The American Society for Anesthetists (ASA) organized a Committee on Standardization of Anesthetic and Resuscitating Equipment, charged with forging a consensus on manufacturing standards for this type of equipment. The ASA, in 1955, approved financial support for the American National Standards Committee Z79 on Anesthesia and Respiratory Equipment to operate under the umbrella of American National Standards Institute (ANSI). Members of this committee included representatives of various medical specialties and principal manufacturers of anesthesia and respiratory equipment. Committee Z79 effectively developed standards for a wide range of respiratory equipment. Under the Medical Device Amendments of 1976, the Food and Drug Administration (FDA) was charged with regulatory responsibility for the safety and efficacy of medical devices. The FDA had a significant impact on refining the standards established by Committee Z79. In 1983, the relationship between ANSI and Committee Z79 was terminated for financial and liability reasons. At that time, the ASA agreed to transfer this committee’s sponsorship to the American Society for Testing Materials (ASTM). The committee’s name was then changed to F29.<sup>236</sup>

Numerous problems are encountered with positive-pressure ventilation. Concern about accidental disconnections from the ventilator led to alarm systems being developed. Adequate humidification of the ventilator air supply had to be ensured.<sup>192</sup> The risk of nosocomial pneumonia was not understood initially. Vigilant attention to sterilization of respiratory equipment, proper suctioning technique, and minimization of stagnant water in tubing and humidification sources was advised to reduce the risk of “ventilator lung.”<sup>198</sup> There was considerable debate for decades over whether translaryngeal intubation was preferred over tracheotomy for patients requiring prolonged mechanical ventilation. Although this subject is still controversial, a consensus conference in 1986 recommended translaryngeal intubation as the preferred initial choice

for obtaining airway control in most patients needing artificial respiration. Secondary tracheotomy could be deferred for up to 20 days or longer depending on the individual situation.<sup>237</sup>

## Weaning

An intriguing problem developed as the use of positive-pressure ventilation became more widespread. Once patients were placed on a mechanical ventilator, how were they to be “weaned” eventually from such respiratory support? This question actually had two components: How could the physician determine when a patient was ready to be weaned? What methods could be used to facilitate the weaning process?

Numerous criteria have been advocated as reasonable indicators that patients are weaning candidates.<sup>238–240</sup> Similarly, numerous techniques have been proposed as useful modalities for maximizing the chances for a successful weaning process. Modifications of ventilator technology in the 1970s led to the proposal of such methods as intermittent mandatory ventilation<sup>241,242</sup> and mandatory minute volume<sup>243</sup> as alternatives to the standard T-piece method of weaning.<sup>244–246</sup> Although many physicians express strong preferences for one weaning modality over another,<sup>247</sup> it has never been shown clearly that modalities such as intermittent mandatory ventilation hasten the weaning process.<sup>248,249</sup> Whether more recently introduced technological advances in ventilator techniques, such as pressure-support and pressure-control ventilation, will improve the weaning process is unclear.

## Conclusion

A rich and complex weave of discoveries in many different scientific and technical areas has brought us to the early 1990s, a time during which physicians have been trained to rely routinely on mechanical ventilation for managing all manner of acute, serious illnesses. Only 40 years ago, cadres of medical and dental students manually ventilated polio patients with respiratory paralysis, and at the turn of the twentieth century, a foot-operated bellows for mechanical ventilation was a remarkable innovation. In the late eighteenth century, the concept of using a bellows and translaryngeal tube for resuscitating the apparently drowned was just being introduced, and in the sixteenth century, Vesalius was forced to make a pilgrimage to the Holy Land to atone for the sin of restarting a Spanish nobleman’s heart by inflating his lungs. The debt we owe to the many pioneers who have contributed to the advances in the field of mechanical ventilation is humbling, just as the hope for future unforeseeable developments in this field is enthralling.

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